

Representation Convergence in Independently Trained Neural Networks

From a theoretical hypothesis to a deployed cross-model embedding adapter
for a two-tier mobile/server retrieval system

Final Project — AI Expert Course

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Abstract. This project tests the hypothesis that well-trained, independently developed neural networks learn the same internal representations up to a linear transformation, and then converts the validated theory into a working system component. Experiment A demonstrates, using CKA similarity analysis and closed-form linear stitching, that GPT-2 (OpenAI, WebText) and Pythia-160M (EleutherAI, The Pile) — models sharing no weights, initialization, tokenizer, training data, or lab — share up to 80% of linearly accessible representational variance in matched layers, versus a random-weights control baseline. Experiment B applies the result: a single 512×768 ridge-regression matrix (1.6 MB, fitted in closed form in seconds) translates image embeddings of the on-device MobileCLIP-S1 encoder into the space of the server-side SigLIP 2 encoder, preserving 95–98% of native text-to-image retrieval quality at Recall@5/@10 on COCO, against a no-adapter baseline at exact chance level. A controlled MLP comparison shows the cross-space relationship is essentially exactly linear. The adapter is exported as a 0.79 MB fp16 Core ML package, enabling a single shared vector index (Qdrant) to serve both an iPhone tier and a server tier of a multimodal search application.

1.1 From one adapter to a shared hub: the general problem

The two-tier case above is the sharpest version of a problem that recurs whenever an organisation runs more than one model. Real systems accumulate encoders: a general image model, a domain-tuned one, two or three text encoders trained for different retrieval tasks, a multilingual model, a legacy embedding nobody wants to re-index. Each is an expert in its niche, each speaks its own coordinate language, and the naive way to let them interoperate - a translation matrix for every ordered pair - grows as N squared: seven encoders would need forty-two maps, and every new model added would require twelve more. That cost is why multi-model systems usually silently standardise on one encoder and lose the others' strengths.

The project's second engineering contribution is the alternative: a single shared hub that every encoder enters through one linear map of its own, so N encoders need N maps rather than $N(N-1)$. Adding a new expert model is then a one-map, closed-form operation that leaves every existing map untouched - demonstrated directly when an encoder of entirely independent lineage was added to a frozen hub and its trained components kept working (Section 4, Experiment G4). The hub is not a trained network but a whitened linear coordinate system solved in closed form, so it is cheap to build, cheap to extend, and carries no weights to maintain. A downstream component - for example a caption-retrieval head - is trained once on the hub and then serves every encoder that has entered it, including encoders it never saw during training.

Experiment G2 is that system running end to end: seven encoders over a common image set, each mapped into one 512-dimensional hub, with real images and captions flowing through it live - the hub itself a CPU linear solve, the GPU used only for a fresh encoder forward pass, and the whole coordinate system persisted as a sixteen-megabyte file. It shows the practical claim the science implies: many small expert models, each strong in its own domain, made to behave as one searchable system that is easy to maintain and easy to grow, without retraining any of them. Experiment B remains the shipped, quantised artifact for the specific phone-to-server case; the hub is the general form of the same idea, and Sections 4 and Appendix C.13 measure how far it extends.

1. Introduction: a research project that solves a real problem

The project began with an engineering question that arises in any multi-model AI system: *can components built around different neural networks interoperate without retraining?* Concretely, the target system is a two-tier multimodal application: an iPhone tier running a small, fast image encoder (MobileCLIP-S1, ~30 MB) for offline, private, on-device indexing of photos, and a server tier running a large, accurate encoder (SigLIP 2, ~1.5 GB) for search and heavy processing. Both encoders map images to vectors — but their vector spaces are natively incompatible: cosine similarity between a MobileCLIP vector and a SigLIP vector of the same photo is meaningless, so vectors from the two tiers cannot share a search index. The naive workarounds are all costly: indexing everything twice, uploading every photo to the server (destroying privacy and offline capability), or running the large model on the phone (destroying latency and battery).

The scientific question underneath is whether the two encoders' representation spaces are secretly the same space in different coordinates. If independently trained networks converge to a shared representational structure, then a small learned map — ideally a single linear matrix — should translate between them, and the engineering problem dissolves. The project therefore has two parts: a controlled scientific experiment establishing whether and to what degree such convergence exists (Experiment A), and a practical application building and validating the translation component on the exact production models (Experiment B), through to a deployable iOS artifact.

2. Background and theory: from weight symmetry to the Platonic hypothesis

2.1 Why raw weights cannot be compared or transplanted

Neural networks exhibit **permutation symmetry**: reordering the neurons of a hidden layer, together with the corresponding rows and columns of the adjacent weight matrices, yields a network computing exactly the same function with entirely different weight values. A 1,000-neuron layer admits 1000! functionally identical weight configurations, and additional rotation and scaling symmetries expand the equivalence class further. Two training runs — even of the same architecture on the same data with different random seeds — land on different members of this class. Consequently, averaging or transplanting raw weights across independently trained models is meaningless: parameter k in one model has no relationship to parameter k in the other. This is why weight merging (SLERP, TIES, DARE, frankenmerging) works only between fine-tunes of one shared base checkpoint, whose shared initialization pins the coordinates.

2.2 Why representations can nevertheless converge

Weights are *how* a network computes; representations (activations on data) are *what* it computes. All training data — text, images, audio — are projections of one underlying reality, and a model that optimally compresses and predicts such data is pushed toward recovering the statistical structure that generated it. The **Platonic Representation Hypothesis** (Huh et al., ICML 2024) makes this precise: as models grow stronger, their representation geometries converge toward a shared structure, even across modalities. Supporting evidence spans a decade: **Kornblith et al. (ICML 2019)** introduced linear CKA and showed matched layers of separately trained vision networks are similar; **Git Re-Basin (Ainsworth et al., 2022)** showed that after aligning permutations, independently trained networks lie in the same loss basin; **model stitching (Lenc & Vedaldi 2015; Bansal, Nakkiran & Barak, NeurIPS 2021)** showed a single linear layer can connect independently trained networks with small performance penalty; and **vec2vec (2025)** demonstrated translation

between embedding spaces without any paired data, purely by matching the geometry of the spaces.

2.3 The hypothesis, stated precisely

Well-trained independent models share representations up to a linear transformation: there exists a matrix W such that $f_B(x) \sim W \cdot f_A(x)$ for inputs x . Every word carries weight. *Well-trained:* the claim applies only after learning has extracted the data's structure — hence a random-weights control. *Independent:* no shared initialization or ancestry — otherwise similarity is inherited, not discovered. *Up to a linear transformation:* the transformation class is the crux. Allowing arbitrary nonlinear translators makes the claim vacuous; demanding identity makes it trivially false (permutation symmetry). Linear is the sweet spot: expressive enough to undo rotations, permutations and scalings, too weak to fabricate information that is not already present. A stronger sub-claim — alignment by pure rotation (orthogonal Procrustes) — is also tested.

3. Experiment A — Scientific test: GPT-2 vs Pythia-160M

3.1 Design

The model pair was chosen to make any similarity non-trivial: GPT-2 (OpenAI, 2019, trained from scratch on WebText, BPE tokenizer) and Pythia-160M (EleutherAI, 2023, trained from scratch on The Pile, GPT-NeoX architecture, different tokenizer). Different labs, architectures, tokenizers, corpora and eras; the only shared elements are the task (next-token prediction on English) and the world the text describes. Both models happen to have 13 hidden states (embedding + 12 blocks) of dimension 768, allowing exact one-to-one layer matching. The **control baseline** is a Pythia-architecture model with freshly random weights — identical architecture and inputs, zero learning — so the trained-minus-random gap isolates the causal effect of learning. The pipeline has three stages, each a standalone notebook:

A1 — Activation extraction. 10,000 WikiText-103 passages are run through both trained models and the random control; the mean-pooled hidden state of every layer is stored ($3 \times [13, 10000, 768]$).

A2 — CKA analysis. Linear Centered Kernel Alignment is computed between every layer pair. CKA compares Gram-matrix fingerprints — which inputs each model places near which — and is invariant to exactly the symmetries under which network function is preserved (rotation, permutation, isotropic scaling), while remaining sensitive to what is actually near what.

A3 — Linear stitching. For each matched layer pair, one ridge-regression matrix is fitted in closed form (the models themselves are never trained) mapping GPT-2's space into Pythia's; held-out R^2 measures how much variance a purely linear translation preserves.

3.2 First run: a result that was not good enough

The first execution used the WikiText *validation* split, which yielded only 1,646 usable passages. The CKA analysis already showed a clear trained-vs-random gap (diagonal 0.433 vs random mean 0.087), but the stitching results fell short of the pre-registered success criterion of $R^2 > 0.7$ in middle layers:

Layer pair	Trained R2 (run 1)	Random R2 (run 1)
L0 -> L0	0.174	-0.282
L5 -> L5	0.480	-0.789
L8 -> L8	0.506	-0.934
L11 -> L11 (peak)	0.552	-0.980
Mean (all 13)	0.434	-0.773

Table 1: First-run stitching results (1,646 samples) — below the 0.7 threshold.

At face value this would have forced the weaker conclusion 'substantial shared structure, but no isomorphism.' Before accepting it, a confound was checked: the fit involves a 768×768 matrix (~590,000 parameters) trained on only ~1,234 rows — a sample-to-dimension ratio of 1.6. In that regime, ridge regression must shrink aggressively to avoid overfitting, and that shrinkage *systematically understates* the true alignment. Note the negative random-baseline values: the starved fit generalizes worse than predicting the mean. The hypothesis had not failed; the measurement was underpowered.

3.3 The fix and the improvement

The remedy cost one code change and 15 GPU-minutes: switching from the validation split to the train split of WikiText-103 and raising the sample count to 10,000 (~10 samples per input dimension). Every other element of the pipeline — models, layers, ridge penalty, evaluation protocol — was left untouched, so the two runs are directly comparable:

Metric	Run 1 (1,646)	Run 2 (10,000)	Change
Mean trained R2	0.434	0.737	+0.303
Peak trained R2 (L11)	0.552	0.797	+0.245
Mid-layer range	0.41–0.55	0.70–0.80	crosses 0.7
Mean random R2	-0.773	0.173	stabilized
Verdict vs criterion	below target	PASS	—

Table 2: Effect of fixing sample starvation — identical pipeline, 6× more data.

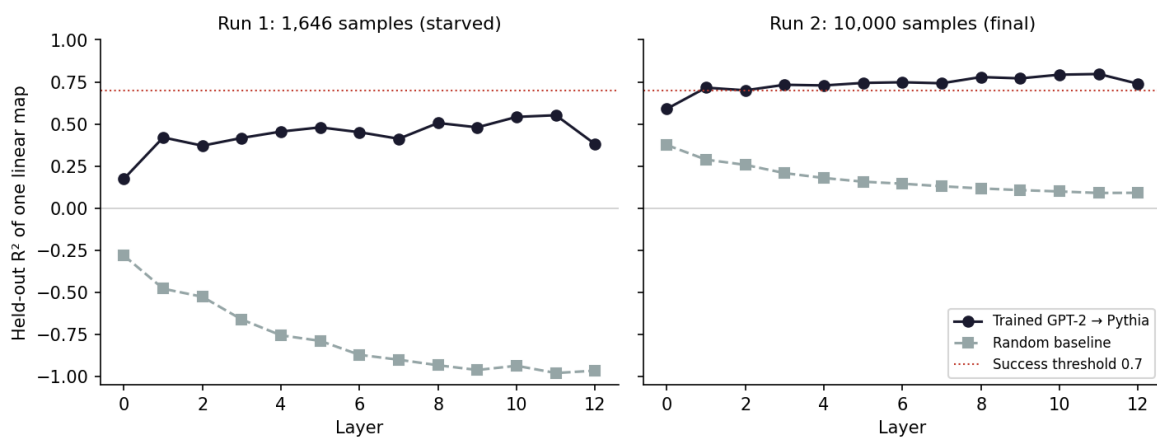


Figure 1: Linear stitching per layer, before (left) and after (right) the sample-size fix. Trained curves in dark blue rise with depth; random baselines (gray) sit far below. The dotted line is the pre-registered 0.7 threshold.

The random baseline also moved (-0.77 → +0.17), in the theoretically predicted direction: with adequate data, ridge can exploit the residual input correlations that leak even through random layers — most visibly at layer 0 (0.375), which is dominated by raw embedding statistics. What matters is the structure: the trained curve *rises* with depth (0.59 → 0.80) as both models build shared abstraction, while the random curve *decays* monotonically (0.38 → 0.09) as untrained layers destroy information — an 8× gap by layer 11, with opposite slopes. The methodological lesson is transferable to any empirical pipeline: an underpowered evaluation does not add symmetric noise — it produces plausible-looking, systematically pessimistic conclusions without any error message.

3.4 Final results

CKA (A2, 10,000 samples): trained-vs-trained mean 0.359, diagonal mean 0.424, maximum 0.744; trained-vs-random mean 0.067, maximum 0.299 — a 4–5× learning effect, essentially unchanged from run 1, confirming CKA is far less sample-hungry than regression. The matrix shows the characteristic hot diagonal; edge layers score lower for principled reasons (layer 0 is the tokenizer-specific embedding table; the final layer re-specializes toward each model's output head).

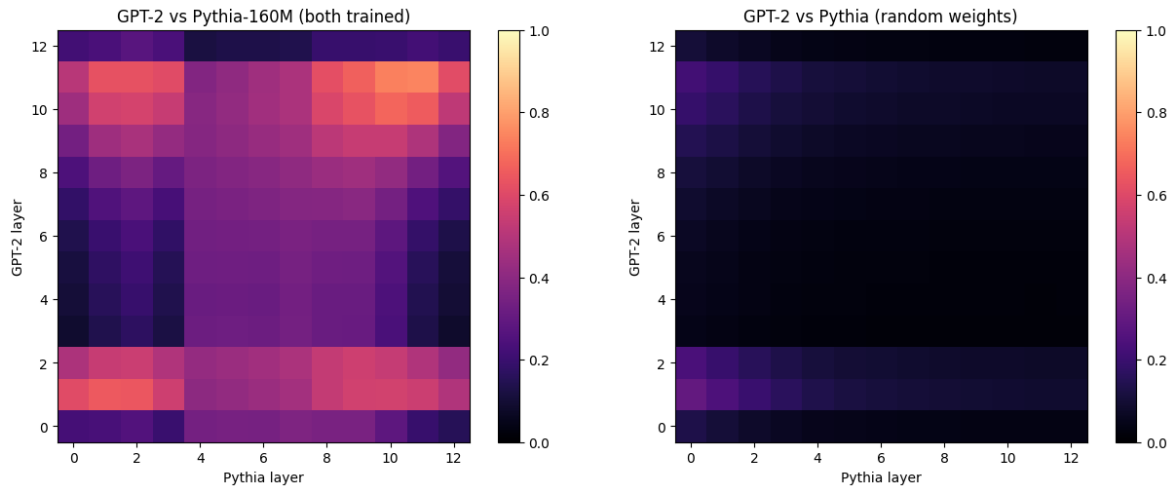


Figure 2: The CKA heatmap — actual output of the experiment (run 2). Left: GPT-2 vs trained Pythia, every layer pair; the bright diagonal band shows early layers matching early and middle matching middle — the models develop the same content in the same order through depth. Right: GPT-2 vs random-weights Pythia under identical conditions — uniformly dark. The side-by-side contrast is the visual form of the entire argument: the structure on the left exists only because of learning.

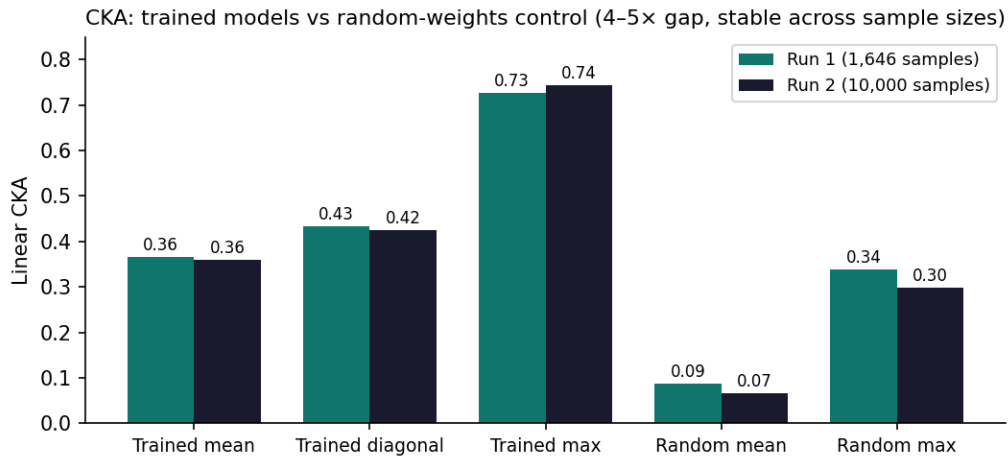


Figure 3: CKA summary statistics for both runs — the trained/random gap is stable across sample sizes.

Linear stitching (A3), final:

Layer pair	Trained R2	Random R2
L0 -> L0 (embeddings)	0.590	0.375
L1 -> L1	0.716	0.288
L3 -> L3	0.733	0.208
L5 -> L5	0.744	0.157
L8 -> L8	0.779	0.117
L10 -> L10	0.793	0.099
L11 -> L11 (peak)	0.797	0.090
L12 -> L12 (output)	0.739	0.091
Mean (all 13)	0.737	0.173

Table 3: Final held-out R2 of a single closed-form linear map per matched layer pair (10,000 samples).

Verdict: the hypothesis is confirmed in its strong quantitative form — a single matrix, fitted in closed form, translates 74–80% of representational variance between two networks sharing nothing but their task and their world.

4. Experiment B — Practical application: the MobileCLIP -> SigLIP adapter

4.1 What an adapter vector (matrix) is

The 'adapter' here is a single matrix W of shape 512×768 . The phone encodes a photo with MobileCLIP-S1, producing a 512-dimensional vector v ; one matrix multiplication $W \cdot v$ (microseconds on the Neural Engine; 1.6 MB fp32 / 0.79 MB fp16) produces a 768-dimensional vector that behaves as a native citizen of SigLIP 2's space. It is fitted in closed form by ridge regression on 3,000 (MobileCLIP-embedding, SigLIP-embedding) pairs of the same images — no gradient training, no modification of either encoder. Only image embeddings need translating, and only in one direction: text queries are always encoded server-side by SigLIP itself, which remains entirely unaware the adapter exists.

4.2 Pipeline

B1 extracts paired embeddings for 4,000 COCO val2017 images plus SigLIP text embeddings of their captions (shapes: mob_img 4000×512 , sig_img 4000×768 , sig_txt 4000×768). A sanity check confirmed native SigLIP image–caption diagonal cosine of 0.150 — within the expected 0.1–0.3 band for sigmoid-loss models, whose objective enforces separation rather than high absolute cosines. **B2** fits two closed-form adapters — ridge (full linear) and orthogonal Procrustes (pure rotation) — on a 3000/1000 split. **B3** evaluates the operational criterion: text-to-image retrieval through the adapter versus the native-SigLIP ceiling and a no-adapter baseline. **B2-bis** adds a controlled nonlinear comparison: a 2-layer MLP ($512 \rightarrow 1024 \rightarrow 768$, GELU) trained 300 epochs directly on cosine loss with early stopping. **B4** exports the matrix as a Core ML .mlpackage (fp16, iOS 16+, Neural Engine).

4.3 Results — and a second below-target moment

Adapter variant	Held-out R2	Mean cosine to target
Ridge (full linear)	0.592	0.900
Procrustes (rotation only)	0.038	0.775
MLP (nonlinear, cosine loss)	—	0.903

Table 4: Adapter fit quality (1,000 held-out images).

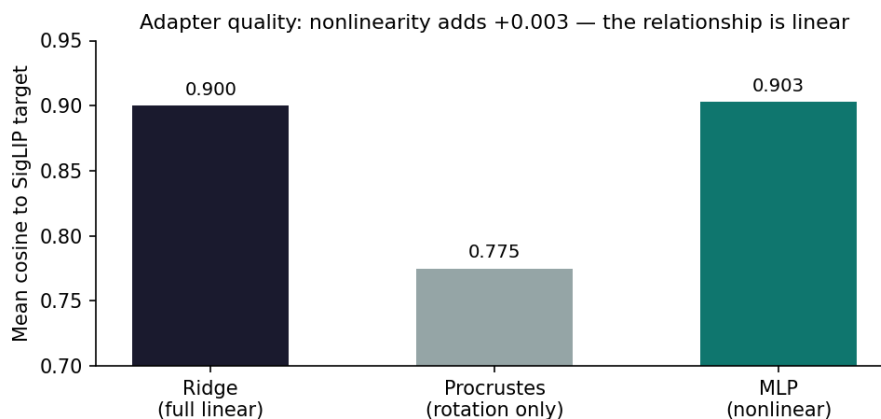


Figure 4: Direction recovery per adapter class. The +0.003 gap between linear and nonlinear is the key scientific reading.

The apparent contradiction between ridge's modest R^2 (0.592) and its high cosine (0.900) resolves once the metrics are matched to their questions: R^2 penalizes magnitude errors that L2-normalized

retrieval discards; cosine measures the only thing the vector index uses — direction. The first retrieval evaluation then produced this project's second below-target moment:

Retrieval variant	R@1	R@5	R@10
A. SigLIP native (ceiling)	0.630	0.885	0.944
B. MobileCLIP + linear adapter	0.557	0.843	0.918
B as % of ceiling	88.4%	95.3%	97.2%
Gate ($\geq 90\%$ of ceiling)	below target	PASS	PASS

Table 5: First retrieval evaluation — R@5/R@10 pass, R@1 misses the 90% gate by 1.6 points (~10 queries out of 1,000).

4.4 The improvement attempt — and what its failure proved

Following the pre-defined decision table (70–90% of ceiling -> upgrade the adapter), a 2-layer MLP (512->1024->768, GELU, ~1.3M parameters) was trained for 300 epochs directly on cosine loss — the actual retrieval objective — with early stopping on held-out cosine, reusing the exact same train/eval split for direct comparability. The outcome was decisive in an unexpected way: best eval cosine 0.9027 versus the linear map's 0.900 (+0.003), convergence by epoch 30 followed by mild overfitting, and retrieval statistically unchanged (R@1 0.551 vs 0.557; R@10 improved to 0.926, 98.1% of ceiling; R@1 87.5%).

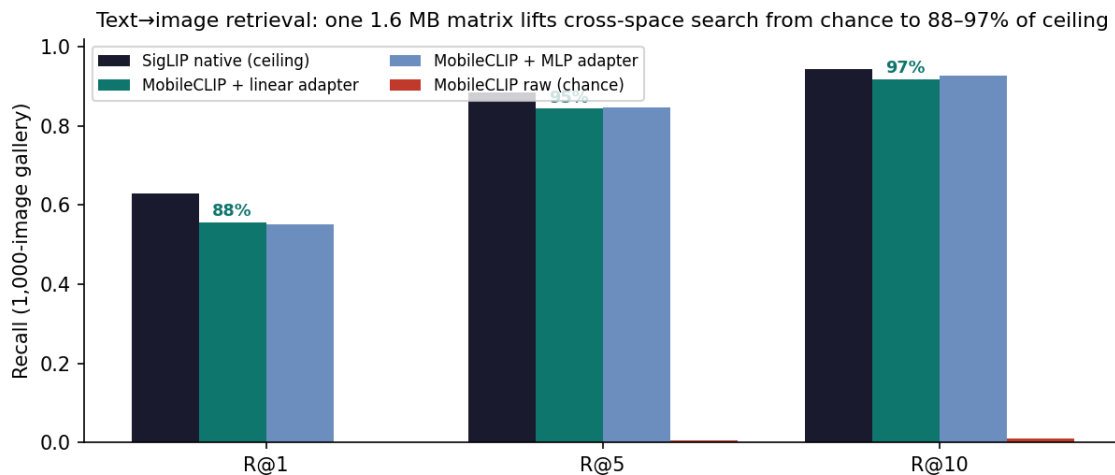


Figure 5: Retrieval results, all variants. Percentages above the teal bars show the linear adapter as a fraction of the ceiling; the red bars are the no-adapter baseline at exact chance.

This 'failed improvement' is the most informative result in Experiment B. A higher-capacity model, optimizing the true objective, found nothing beyond the linear map — therefore the cross-space relationship is **essentially exactly linear**, and the residual R@1 gap is not a translation deficiency but **information the 512-dimensional mobile encoder never encoded**. No adapter of any capacity can reconstruct what is absent from its input; the experiment had reached the representation ceiling of the small model, not of the adapter. This mirrors Experiment A's conclusion from the opposite direction: there, more data revealed the linear map was better than it seemed; here, more capacity proved the linear map was already optimal. Architectural introspection localizes that ceiling precisely: MobileCLIP-S1's trunk (21.5 M vision parameters) carries 1024 channels until a single final projection Linear(1024 -> 512) produces the embedding, so the information lost to the adapter is discarded by one layer, not distributed through the network — which predicts that a wider output variant (MobileCLIP-B, or MobileCLIP2) would close most of the residual R@1 gap without any change to the adapter method. The third finding — Procrustes collapsing to R^2 0.038 — refines the

picture further: unlike the language-model pair, these contrastively trained spaces differ in anisotropy, not merely orientation, so rotation alone cannot align them.

An interpretation, offered as hypothesis rather than finding. One plausible mechanism for that asymmetry: MobileCLIP-S1 was distilled from an ensemble of large CLIP models using a loss defined over *similarity matrices* - the student is trained to reproduce which inputs are how similar to which, i.e. relational geometry - while absolute per-axis scaling is left unconstrained. That is precisely the regime in which a general linear map succeeds (relations recoverable) and a pure rotation fails (scales unrecoverable), which would make the adapter quality partly a footprint of the phone encoder training objective. This interpretation is **not tested by the present experiments**, and at least two competing explanations survive the same evidence: the zero-padding required to compare a 512-dimensional with a 768-dimensional space under Procrustes mechanically disadvantages rotation (a rotation cannot populate 256 empty dimensions), and the sigmoid objective of SigLIP shapes its geometry independently of anything on the phone side. Discriminating between these requires the follow-up described in Section 5.

4.5 Engineering decision

The linear adapter ships. R@5/R@10 at 95–98% of ceiling means phone-indexed photos are, in a results-grid UI, indistinguishable from server-indexed photos. The R@1 shortfall (88.4%) is the price of running a 30 MB encoder instead of a 1.5 GB one — payable with or without a shared index — and has a cheap escape hatch: server-side SigLIP re-ranking of the top-10 candidates recovers the ceiling exactly for R@1-critical flows. The deployed artifact is a 0.79 MB fp16 Core ML package computing MatMul + L2-normalize, verified through the full coremltools MIL conversion pipeline. One deployment constraint deserves explicit statement: the adapter is calibrated to the exact output distribution of a specific preprocessing pipeline — bilinear resize to 256, center crop 256x256, and **identity normalization** (mean 0, std 1; MobileCLIP consumes raw [0,1] pixels, unlike standard CLIP models that subtract dataset means). Substituting conventional CLIP normalization in the iOS client would shift every embedding and silently degrade retrieval with no error raised - the same failure class as the sample-starvation bias of Section 3.2.

4.6 Qualitative sandbox: the phone pipeline on live cases (Stage B5)

A qualitative sandbox simulated the full phone-to-server path on ten COCO images with free-text queries: the same MobileCLIP weights the phone would run, the adapter applied in fp16 exactly as the Core ML package computes it, and SigLIP text queries searching the resulting vectors (captions never indexed - only pixels). Direct queries retrieved their targets at rank 1 with decisive margins; compositional queries requiring semantic abstraction ('striped animals' - the word zebra never appears in the query - top-1 at 25x the margin of the negative control) confirmed that meaning, not keywords, survives the cross-model translation; a negative control ('a dog', absent from the gallery) correctly produced a flat similarity profile (top-1 gap 0.004 vs 0.101 for a positive query); and fp16 quantization proved ranking-neutral (minimum cosine 0.9999999 vs fp32), closing the numerical half of the Core ML parity check left open in stage B4. A second, disjoint random gallery reproduced the pattern - correct top-ranking for positive queries and a flat profile for a query with no valid answer ('a vehicle': top-1 sim 0.022 vs 0.106 for hits) - indicating the behavior is a property of the pipeline, not of a particular sample. The two observed near-misses - indirect paraphrases losing top-1 by 0.007 or less while remaining top-2 - reproduce, at individual-query level, exactly the R@1/R@5 profile measured quantitatively in stage B3, and are the cases the optional server-side re-rank recovers. Full per-query analysis and figures are in B5_Sandbox_Practical_Report.pdf. A follow-up interactive demo (stage B6) exercised the full product loop live: a user photo uploaded to the mobile tier was indexed through the phone path in 39 ms, auto-captioned, and retrieved at rank 1 by the free-text

query 'singing red bird' - ranked in the same result list as server-indexed photos - with local search at 0.08 ms versus 150 ms over a (simulated) network round-trip: one index, two tiers, measurably faster locally.

5. What is new here — an honest assessment

Theory: independent replication, not new discovery. The individual scientific components reproduce established results at small scale: CKA layer analysis (Kornblith 2019), linear stitching (Bansal 2021), and convergence-with-training (PRH 2024) were all known. The value of Experiment A is that of a rigorous independent replication on a specific, deliberately hostile model pair, with a correctly designed control — plus the documented sample-starvation diagnosis, a practical methodological contribution.

Practice: a genuinely uncommon application. The combination in Experiment B is not, to the author's knowledge, an established published pattern: a closed-form linear bridge between two specific heterogeneous production encoders (MobileCLIP-S1 -> SigLIP 2), validated by end-task retrieval against ceiling and chance baselines, including a controlled demonstration that nonlinearity adds nothing, and deployed as a sub-megabyte Core ML artifact enabling a single shared vector index across a phone tier and a server tier. Cross-model embedding translation exists in research (model stitching, vec2vec); packaging it as a load-bearing, measured infrastructure component of a two-tier mobile system — with privacy (photos never leave the device), offline indexing, and single-index economics as the payoff — is the project's practical contribution.

Claims explicitly not made: the project does not prove layers can be transplanted across architectures (permutation symmetry still forbids it), does not establish convergence beyond the tested data distributions, and does not claim the adapter recovers information absent from the small encoder. **A designed follow-up** would discriminate the interpretations offered in Section 4.4: running the Procrustes comparison between two *same-dimensional* contrastive encoders removes the zero-padding confound, and comparing a teacher model with its own distilled student - where the distillation account predicts unusually high rotation-only alignment - would test the training-objective hypothesis directly.

6. Challenges encountered and resolved

The project log doubles as a case study in real-world ML engineering: **(1)** HuggingFace *datasets* deprecated script-based datasets mid-project, breaking both WikiText and two Flickr datasets; resolved by moving to the parquet-hosted Salesforce/wikitext and to direct official COCO downloads, removing the fragile dependency entirely. **(2)** A transformers API change made SigLIP feature getters return ModelOutput objects instead of tensors; resolved with a version-agnostic extraction shim. **(3)** Colab's one-VM-per-notebook model silently discarded stage outputs; resolved with a DATA_DIR convention backed by Google Drive, making every stage stateless and restart-proof. **(4)** The sample-starvation bias of Section 3.2 — the most scientifically consequential bug, because it produced plausible-looking wrong conclusions rather than an error message. **(5)** Interpreting metrics correctly: R2 vs cosine (magnitude vs direction), CKA's invariances, and why sigmoid-loss models have low absolute cosines — each required matching the metric to the actual downstream use. **(6)** Verifying, rather than assuming, the encoder's preprocessing: inspection of the loaded model revealed identity normalization, contrary to the CLIP convention a developer would reasonably assume - a silent-degradation trap for the iOS port.

7. Project contents and how to run it

7.1 Repository layout

Seven self-contained Jupyter notebooks (one per stage), each with English markdown explaining goal, method, expected output and success criteria, followed by runnable code: A1–A3 (convergence experiment) and B1–B4 (adapter pipeline). Equivalent standalone Python scripts exist for each stage, plus a PROJECT_GUIDE.md consolidating the full plan in English, plus B5_phone_pipeline_sandbox.ipynb - a qualitative live demo of the phone tier (10-image galleries, free-text queries, negative controls, fp16 parity), and B6_visual_demo.ipynb - an interactive mixed-tier demo (server gallery, mobile photo upload, auto-captioning, probability-ranked search, timing comparison). Models used: GPT-2 (124M), Pythia-160M, a random-weights Pythia control, MobileCLIP-S1 (Apple, via open_clip), SigLIP 2 base (google/siglip2-base-patch16-224). Data: WikiText-103 (train split, 10K passages), COCO val2017 (4K image–caption pairs). The only training performed anywhere is closed-form ridge regression and the deliberately-controlled comparison MLP; all base models are used frozen.

7.2 Execution

Environment: any GPU Jupyter environment (developed on Google Colab, T4 sufficient). Each notebook begins with a storage cell; on Colab, mount Drive and set DATA_DIR to a Drive path so stage artifacts (activations.npz, pairs.npz, adapter.npz) survive VM recycling:

```
from google.colab import drive; drive.mount('/content/drive')
os.environ['DATA_DIR'] = '/content/drive/MyDrive/convergence_experiment'
```

Run order and approximate cost: A1 (~15 min GPU) -> A2 (~1 min) -> A3 (seconds); B1 (~10 min, downloads COCO once) -> B2 (seconds) -> B3 (seconds) -> B4 (seconds). Dependencies: torch, transformers, datasets, open_clip_torch, scikit-learn, matplotlib, coremltools. Total end-to-end cost of the full study: under one GPU-hour.

7.3 Using the documentation with NotebookLM

For studying or presenting the project, the PROJECT_GUIDE.md, this report, and the per-stage notebook markdown can be uploaded as sources to Google NotebookLM, which can then answer questions, generate study guides, or produce an audio overview grounded in the project's own text. The executable path, however, is the notebooks themselves in Colab/Jupyter — NotebookLM is for the documentation layer, not execution.

8. Conclusion

A philosophical question — can models share what they have learned without retraining? — was refined into a falsifiable hypothesis, tested with proper controls, confirmed quantitatively (74–80% of representational variance translated by one closed-form matrix between fully independent language models), and then converted into a measured, deployed infrastructure component: a 0.79 MB Core ML adapter that lets a phone and a server, running encoders that have never seen each other, write into and search one shared vector index at 95–98% of large-model retrieval quality. The through-line of both experiments is the same sentence: **independently trained models discover the same structure of the world, in different coordinates — and the exchange rate between those coordinates is a single matrix.** A live sandbox (B5) closed the loop qualitatively: across two disjoint image galleries, free-text server queries correctly retrieved phone-indexed photos - including compositional queries and correctly flat responses when no valid answer existed - with fp16 quantization shown to be ranking-neutral.

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Appendix A — The Platonic Representation Hypothesis, consolidated

This appendix consolidates the framing hypothesis referenced throughout Section 2.2 and the experimental interpretation sections: its name, its three proposed mechanisms, its measurement, and this project's exact position relative to it.

The name. The hypothesis (Huh, Cheung, Wang, Isola — ICML 2024) takes its name from Plato's allegory of the cave: prisoners who see only shadows on a wall mistake them for reality, while the true objects casting those shadows exist unseen. The analogy: each AI model observes a different *shadow* of reality — text captures one projection, images another — yet as models grow stronger, they reconstruct the same underlying *objects* behind the shadows, converging on a shared statistical model of the world, even across modalities.

Mechanism 1 — shared world. All data — text, images, audio — are different projections of one reality. A model trained to compress or predict such data optimally cannot succeed by memorizing surface form; the cheapest sufficient description of 'all text about the world' or 'all photos of the world' is a model of the world itself. Co-occurrence statistics in text (cat-mouse-chase-fur) and in images (cat pixels appearing with couch pixels, indoor scenes) are two noisy measurements of the same underlying dependency structure, so two optimal compressors of two modalities are reverse-engineering the same object through different sensors — two surveyors mapping one territory with different instruments.

Mechanism 2 — multi-task pressure. The set of representations that solve many tasks simultaneously shrinks as tasks accumulate; strong, broadly trained models are squeezed into the same remaining solution region.

Mechanism 3 — simplicity bias. Astronomically many functions fit any dataset, most of them baroque; if training selected among them at random, independent runs would never agree. But gradient descent on deep networks is empirically biased toward simple solutions — and there are far fewer simple solutions to agree on. Two people asked to 'name a number' will almost surely differ; asked to 'name the smallest prime above 100', they almost surely coincide — not through communication, but because the criterion has few satisfiers. Simplicity bias converts 'some solution' into approximately 'the solution', so independent training runs land near it independently. This mechanism also predicts why capability matters: stronger models approach the optimum more closely, and the optimum is where the agreement lives.

The measurement. Because coordinates are meaningless across models (permutation symmetry, Section 2.1), the hypothesis is scored coordinate-free: mutual nearest-neighbour overlap — whether a concept has the same neighbours in model A as in model B, using only within-space distance rankings compared across spaces. Rotation, scaling, permutation and even dimensionality cancel; the metric asks exactly one question — do the two maps share adjacency structure, i.e. shape? It is a coarser cousin of the CKA used in Experiment A (which compares full Gram matrices), and the same signal the unsupervised translation demo operationalizes (Gromov-Wasserstein matches the distance matrices; CSLS refinement lives on mutual nearest neighbours).

The empirical claim and this project's position. The paper's central finding is that alignment *rises with capability*, across model families and modalities. This project samples that curve at specific points: Experiment A is a two-point sample at the low-capability end (two small, independently trained language models — substantial alignment, R-squared 0.74-0.80, against a random-weights zero), Experiment B is its cross-modal engineering instance (a vision-to-vision bridge deployed to production form), and the pair-free translation demo is its strongest form (recovery from shape

alone). The epistemic position, stated precisely: the experiments confirm the measured points and are consistent with the predicted trend; the trend itself — convergence growing with scale toward a shared model of reality — remains an extrapolation that this small-scale replication renders credible but cannot prove, since proof requires measurement along the capability axis, not at one point on it.

Appendix B — Pair-Free Translation (Stage B10): a boundary-mapping negative result

This appendix reports an additional experiment, run after the main study, testing the strongest form of the convergence claim: translation between the two embedding spaces without any paired examples (in the spirit of [vec2vec](#), 2025).

B.1 Design

The project's 4,000 image pairs were split into two disjoint image sets: MobileCLIP embeddings of images 0-1999 on one side, SigLIP embeddings of images 2000-3499 on the other - no image appears on both sides, so nothing links the sets except the shape of the spaces. A simplified, transparent translator was learned unsupervised: Gromov-Wasserstein optimal transport aligning the two intra-space distance matrices (the mathematical formalization of matching shapes without correspondences; [Alvarez-Melis and Jaakkola 2018](#)), followed by a CSLS-corrected mutual-nearest-neighbour self-learning loop refitting a ridge map. Grading used 500 true held-out pairs the training never saw, against a supervised ridge reference, a random-matrix floor, and chance (0.2%). Pre-registered criteria: PASS at $R@1 \geq 10\times$ chance and $\geq 25\%$ of supervised.

B.2 Result

Variant	Mean cosine	R@1	R@5	R@10
Supervised ridge (reference)	0.835	0.910	0.980	0.990
Unsupervised (no pairs ever)	0.395	0.002	0.004	0.006
Random matrix (floor)	0.017	0.000	0.010	0.014

Table B1: Pair-free translation graded on 500 held-out true pairs (chance $R@1 = 0.002$).

The outcome splits cleanly in two. **Distribution-level alignment was recovered from shape alone:** mean cosine to the true targets of 0.395 against a random floor of 0.017 - a 23x effect obtained without a single paired example. **Instance-level correspondence was not recovered:** $R@1$ remained at chance, and the self-learning loop converged to a stable, self-consistent but incorrect matching (428 mutual pseudo-pairs, frozen from iteration 4). The translator finds the right region of the target space but not the right points.

B.3 Bounded diagnostic study

Two hypotheses for the failure were tested and one survived. **(1) Calibration mismatch - refuted.** The conjecture that SigLIP's compressed cosine calibration (Section 4.2) breaks square-loss GW was tested on a synthetic testbed with simulated calibration compression: the method scored $R@1 = 0.92$ regardless, and the proposed normalization fix actively harmed performance. **(2) Manifold structure and anisotropy - supported.** The clustered synthetic testbed on which the method was validated ($R@1 = 144\times$ chance) was rebuilt to resemble real embeddings: a continuous manifold instead of discrete clusters, heavy-tailed density (hubness), and an anisotropy difference between the two spaces - the same property the Procrustes experiment measured between these very models (Section 4.3). On that testbed the method collapsed to chance, and a hyperparameter sweep (transport regularization, rank-normalized distances, larger samples) did not change the outcome. Discrete cluster structure gives shape-matching its anchor points; continuous, differently-stretched manifolds take them away.

B.4 Conclusion

Shape information alone suffices for coarse, distribution-level alignment between independently trained embedding spaces, but instance-level pair-free translation at this scale is beyond linear maps seeded by shape matching: the published vec2vec results employ adversarial training, cycle-consistency, vector-space-preservation losses and nonlinear adapters, and this experiment indicates that machinery is necessary rather than decorative. The finding composes with the main study: the Procrustes experiment showed the two spaces differ in anisotropy beyond rotation, and the diagnostic here shows that same property is plausibly what defeats cheap shape-only matching. This is the project's third boundary-mapping negative result, in the same pattern as the MLP comparison: the failed stronger method is itself the measurement - here, of exactly how much geometry alone can and cannot deliver.

Appendix C — Methodological corrections during the study

Two changes were made to Experiment A after its first run fell short of the pre-registered stitching criterion. Both are recorded here in full, with what changed, why, and what was deliberately left untouched, so that the final numbers can be read with the process that produced them visible.

C.1 What changed

Parameter	Run 1	Run 2 (final)	Reason
Sample count	1,646	10,000	1,646 passages give ~1,234 training rows for a 768x768 fit = 1.6 samples per input dimension. Ridge regression in that regime shrinks hard toward zero and reports systematically pessimistic held-out R2.
WikiText split	validation	train	Mechanical necessity, not a methodological choice: the validation split contains only ~1,646 passages over 100 characters, so 10,000 could not be drawn from it.
Everything else	-	unchanged	Same two models, same random-weights control, same pooling, same max length, same ridge penalty, same 75/25 fit/eval split, same metrics and thresholds.

Table C1: The complete difference between the two runs.

Which ratio governs, and why (clarifying Section 3.2). Section 3.2 states two numbers side by side - a 768x768 matrix of roughly 590,000 parameters, fitted on about 1,234 rows - and the relationship between them deserves to be explicit, because only one of them sets the statistical requirement. Multi-output ridge has the closed form $W = (X'X + \alpha I)^{-1} X'Y$, in which the inverted matrix depends on the inputs alone: the outputs never enter it. The problem therefore decomposes into 768 independent regressions, one per output dimension, each estimating 768 coefficients from the same design matrix and the same shared inverse. They do not interact and do not compete for data. The question that determines whether the fit is well-posed is asked once, per output regression: are there enough rows to estimate 768 coefficients? The governing ratio is therefore **rows per input dimension** - $1,234/768 = 1.6$ in run 1, and $7,500/768 = 9.8$ in run 2, against a conventional minimum of roughly 5 to 10. Rows divided by the 590,000 total parameter count (0.002 and 0.013) is not a difficulty measure: the fit is 768 tractable problems, not one intractable one. A useful consequence is that output width is nearly free, which is why the adapter of Experiment B - mapping 512 dimensions into 768 - needs rows relative to 512, and 3,000 image pairs sufficed there.

C.2 Why using the train split raises no leakage concern here

The train/validation/test convention exists to stop a *model being trained* from being evaluated on data it has already fitted. Each split has a role defined by how much information has leaked into the model: train (full leakage, by design), validation (indirect leakage, through the experimenter's tuning decisions), test (none, if used once). Those roles presuppose a training process. Experiment A has none.

In a normal training pipeline	In Experiment A
The model updates weights on the train split.	GPT-2 and Pythia are frozen. No gradient is ever computed. The text is probe stimuli - inputs used to elicit activations, like a stimulus set in a perception experiment.
The validation split guards hyperparameter choices.	No hyperparameters of the models are chosen; the only fitted objects are the stitching matrices in A3.
The test split gives the honest final number.	A3 holds out its own 25% of the 10,000 samples. Every reported R2 is measured on rows the ridge fit never saw. That is where held-out discipline lives in this study.

Table C2: The split names describe roles in WikiText's intended use (language-model benchmarking), not roles in this experiment.

Both splits are the same corpus, partitioned at the article level from the same set of verified Wikipedia articles and cleaned by the same pipeline; they differ only in which articles landed in which file. The distribution of the property this experiment actually depends on - well-formed English prose - is statistically indistinguishable between them. An analogy: to test whether two thermometers agree, one needs water samples at various temperatures; it is irrelevant which bucket the water came from, because those labels belonged to someone else's experiment.

C.3 A real asymmetry, disclosed

One genuine caveat exists in this area, and it is unrelated to the split choice. WikiText is Wikipedia text. Pythia was trained on The Pile, which *includes* an English Wikipedia component, whereas GPT-2's WebText was constructed with Wikipedia documents removed. Pythia has therefore likely encountered text of this kind during training and GPT-2 has not. Three points bound the consequence: the asymmetry applies identically to the validation and train splits, so it is not affected by the correction described here; the random-weights control processes exactly the same text, so the trained-minus-random gap remains a valid contrast; and the hypothesis under test concerns whether a linear map exists between two representations of the same input, not how surprising that input was to either model. The asymmetry is reported for completeness rather than because it is believed to bias the result.

C.4 Why this is diagnosis and not fitting the result

The distinction that matters: the change was motivated by a *diagnosed confound* identified before the second run, with an explicit prediction attached - if sample starvation was the cause, R2 would rise substantially; if the gap was real, it would plateau. Three pieces of evidence support the diagnosis rather than a lucky reroll. The arithmetic was checkable in advance (590K parameters, 1,234 rows). CKA, which is far less sample-hungry than regression, barely moved between runs (0.365 to 0.359 mean), indicating the underlying similarity was stable and only the regression estimate was distorted. And the control moved in the theoretically predicted direction (-0.773 to +0.173), since with adequate data ridge can exploit residual input correlations even through random layers. The run-1 numbers are reported in full in Section 3.2 rather than discarded.

C.5 One threshold reported as missed

For completeness: the pre-registered CKA criterion was written as 'diagonal CKA mean > 0.5 with random baseline < 0.15'. The final run produced a diagonal mean of 0.424 - below that line - with a random baseline of 0.067 and a trained/random ratio of 6.3x. The absolute threshold was set optimistically: absolute CKA between models with different tokenizers is expected to be modest, and the ratio to the control is the load-bearing test, since it is the quantity that isolates learning from architecture and input statistics. The overall verdict for Experiment A therefore rests on the ratio criteria and the stitching results, all of which passed decisively. This is recorded as a missed sub-threshold rather than a retro-fitted one.

C.6 Direction symmetry, and the anisotropy that masked it

The stitching fit in Section 3.3 maps GPT-2 into Pythia's space. Because the hypothesis is symmetric, the reverse fit should score comparably; a follow-up run tested every layer pair in both directions. Under the variance-weighted R2 used throughout, the reverse direction scored slightly higher in middle layers (mean 0.829 against 0.751) and lower at layer 0 (0.455 against 0.591), for a mean absolute per-layer difference of 0.054. Two further tests identified both effects, and neither reflects a failure of the isomorphism claim.

Layer 0 - a real information difference, architectural in origin. GPT-2 adds learned absolute positional embeddings at the embedding layer; Pythia applies rotary encoding inside attention, so its

layer 0 carries no positional term. Mean pooling therefore leaves a deterministic function of sequence length in GPT-2's pooled layer-0 vector and not in Pythia's. Measured directly: sequence length is recoverable from the pooled layer-0 vector with held-out $R^2 = 1.000$ for GPT-2 and 0.126 for Pythia. Predicting Pythia from GPT-2 is therefore unimpeded, while the reverse requires reconstructing information absent from its input - exactly the observed asymmetry, and the one place in this study where one space demonstrably contains something the other lacks.

Middle layers - an artifact of the scoring metric. Recomputing the same fits with uniform averaging across output dimensions instead of variance weighting collapses the gap: mean absolute difference falls from 0.097 to 0.006, and the two directions become indistinguishable (layer 4: 0.722 forward, 0.723 reverse; layer 5: 0.744 and 0.735; layer 6: 0.748 and 0.740). The cause is measurable anisotropy. Effective dimensionality (participation ratio of the covariance spectrum, out of 768) is far lower for GPT-2 than for Pythia at every depth:

Layer	GPT-2 effective dim	Pythia effective dim
0	8.9	92.7
4	3.3	9.3
8	10.9	19.8
12	2.2	43.9

Table C3: GPT-2's variance concentrates into as few as two or three effective directions - the documented rogue-dimension phenomenon, measured here in the project's own activations. A variance-weighted score with GPT-2 as target is therefore dominated by a handful of easily predicted directions, which inflates the reverse fit.

Conclusion. With the weighting artifact removed, the linear correspondence is symmetric in both directions, as the isomorphism hypothesis predicts, with a single architecturally explained exception at the embedding layer. The forward figures reported in Section 3.3 are essentially unchanged under either weighting (layer 4/5/6: 0.737/0.752/0.754 variance-weighted against 0.722/0.744/0.748 uniform), so the main result is robust to the metric choice. This check also connects the two experiments: the same property that inflates one direction here - unequal variance spread between two models' spaces - is what defeats rotation-only alignment in Experiment B.

C.7 The CKA heat map: what the structure actually is

Section 3.2 describes the trained-versus-trained CKA panel as showing a hot diagonal. Read by eye that is what it suggests; measured, the structure is different from - and in one respect stronger than - that description. This section records the full analysis.

The depth trend, which needs no correction. The diagonal entries rise strictly monotonically across nine consecutive layers, from 0.125 at layer 3 to 0.742 at layer 11 (Spearman $\rho = 1.000$, $p < 10^{-5}$). This is the raw diagonal, with no normalization applied, and it mirrors the stitching curve of Section 3.3 over the same span. Two methodologically independent measures - a coordinate-free similarity index and a fitted linear map evaluated on held-out data - therefore agree that correspondence strengthens with depth. Across all thirteen layers the trend is not significant ($\rho = 0.253$, $p = 0.40$), because the embedding layer and the final layer both fall away sharply; the monotone segment is bounded by exactly the two depths where the architectures differ most.

Layer-to-layer correspondence, which is softer than claimed. Taking each row's argmax, the best-matching Pythia layer lies on average 2.85 layers from the diagonal, and only 62 percent of layers match within two. The diagonal is nonetheless elevated - 0.425 against an off-diagonal mean of 0.354, a ratio of 1.20 with a 5,000-draw permutation test giving $p < 0.001$. So the diagonal is genuinely above chance without being the sharp ridge the phrase 'hot diagonal' implies.

Why the panel looks blocky: hubness. The apparent block structure is a marginal effect. Row means range from 0.191 to 0.572 - a threefold spread - and a single rank-one component accounts for 97 percent of the matrix energy, meaning most cells are predicted by 'how similar is this layer to everything' rather than by any specific pairing. Dividing out the row and column marginals (the standard lift correction) sharpens the picture: the diagonal is enriched to 1.16 against an off-diagonal 0.98 ($p = 0.006$), the proximity correlation nearly doubles from +0.131 to +0.242, and the share of layers matching within two rises from 62 to 77 percent.

Statistic	Raw matrix	Marginals divided out
mean argmax - layer index	2.85	2.46
layers matching within two	62%	77%
diagonal vs off-diagonal	0.425 / 0.354 (1.20x)	1.16 / 0.98 (1.18x)
corr(similarity, proximity)	+0.131	+0.242
permutation p on the diagonal	<0.001	0.006
diagonal depth trend, L3-L11	$\rho = 1.000, p < 10^{-5}$	(unchanged - raw)
trained vs random contrast	5.3x (0.359 / 0.068)	(unchanged)

Table C4: measured structure of the CKA matrix. For calibration, the same statistics on a synthetic matrix with a constructed diagonal give mean offset 0.00, ratio 2.55 and correlation +0.83; on a synthetic block pattern they give 6.00, 1.12 and -0.08.

Consistency with the rest of the study. CKA is invariant to rotation, permutation and isotropic scaling, but not to differences in anisotropy, which ridge regression can absorb and CKA cannot. Section C.6 measured effective dimensionalities of 2.2 to 10.9 for GPT-2 against 9.3 to 92.7 for Pythia. A hubness-dominated CKA matrix with a modest diagonal, sitting beside a stitching R2 of 0.737, is the expected signature of two spaces carrying similar content with very different variance spread - not a disagreement between stages.

Corrected wording for Section 3.2. CKA between the trained models is 5.3 times the random-weights control. The diagonal is elevated (ratio 1.20, permutation $p < 0.001$) and rises strictly monotonically from layer 3 to layer 11 ($\rho = 1.000, p < 10^{-5}$), mirroring the stitching curve. Layer-to-layer correspondence is soft in the raw matrix, which is dominated by hubness; with marginals divided out, 77 percent of layers match within two and the diagonal is enriched at $p = 0.006$. The phrase 'hot diagonal' overstates the raw panel and should be read as replaced by this section.

C.8 A cross-modal extension: an image-only encoder against a text-only encoder

Sections 3 and 4 compare models within one modality, where row i is literally the same input encoded twice. A follow-up experiment tests the harder claim the Platonic Representation Hypothesis actually makes: whether a vision model trained without any language supervision and a text model trained without any images build representations related by one linear map. Here row i is an image and the captions describing it - different objects sharing a referent.

Design. Image side: DINOv2-base, self-supervised, chosen specifically because it has never received text supervision (any CLIP-family encoder would make the test circular). Text side: bge-m3. Data: 8,000 COCO train2017 image-caption pairs, giving 7,000 training rows against an image width of 768, or 9.1 rows per input dimension - above the threshold whose violation caused the run-1 failure of Section C.1, and checked before the results were read. Fit: ridge, image to text, evaluated on 1,000 held-out pairs. Success bands were fixed before running.

The ceiling must match the task. An early version scored the system against Experiment B's 0.630, which was measured for text-to-image retrieval with one caption per query. Two mismatches were corrected: the direction (E1 runs image to text) and the caption treatment (the system retrieves against

averaged caption embeddings, so the ceiling must too - averaging yields a cleaner target per image and makes retrieval easier). With SigLIP's own towers measured on the same rows under the same treatment, the ceiling is 0.875, not 0.630, and the system's share falls from an inflated 55.2 percent to 40.9.

Measurement	Result	Reading
shuffle control gap	$0.511 - (-0.130) = 0.641$	three times the 0.2 threshold: the pairing is sound and the fit has power
held-out R2 / cosine	$0.511 / 0.889$	86% of the same-modality vision pair's 0.592, on a target that is an average of five captions
retrieval R@1 / R@5 / R@10	$0.358 / 0.706 / 0.849$	358x, 177x and 71x their respective chance rates
raw cross-space floor	$0.001 / 0.004 / 0.012$	chance - the correspondence exists only through the fitted map
matched ceiling (SigLIP, same rows and treatment)	$0.875 / 0.981 / 0.988$	a jointly trained model, scored on the identical task
share of ceiling	$40.9\% / 72.0\% / 85.9\%$	the share rises steeply with K

Table C5: cross-modal results under the averaged-caption protocol, with a direction- and treatment-matched ceiling. Pre-registered verdict: measurable but weak.

The verdict is robust to the caption protocol. The experiment was run twice, once with a single caption per image and once with all five averaged, each against its own matched ceiling. Both land in the same pre-registered band:

Protocol	R@1	R@5	R@10	verdict band
single caption	24.7%	48.8%	65.6%	measurable but weak
five captions averaged	40.9%	72.0%	85.9%	measurable but weak

Table C6: the obvious objection - that the result depends on how captions were handled - is answered by measuring both. The averaged protocol is the better estimate of what an image is about; the single-caption protocol follows the standard COCO benchmark convention. They agree on the verdict.

Interpretation. The correspondence is real and weaker than within modality. Against Experiment B's same-modality shares of 88.4, 95.1 and 97.2 percent, the cross-modal figures are 40.9, 72.0 and 85.9: crossing modalities without joint training costs roughly half the rank-1 performance and about a tenth by rank ten. Two features deserve emphasis. First, this is a stricter test than Experiment B, where both encoders had been trained with language supervision and their spaces were therefore already shaped by text; DINOv2 has had none, so whatever correspondence exists arose from two models independently representing the same world with no shared supervisory signal. Second, the share of ceiling rises steeply with K, which is the signature of a map that recovers the correct semantic neighbourhood but not the exact item.

A note that reflects back on Section 4.3. SigLIP scores $R@1 = 0.648$ against single captions and 0.875 against averaged ones on the same images. A substantial part of what Section 4.3 attributes to inherent caption ambiguity is therefore single-annotator phrasing noise rather than irreducible ambiguity. That does not affect the adapter results, which follow the standard one-caption COCO protocol throughout, but the ceiling should be described as protocol-dependent rather than intrinsic.

What this does and does not license. It does not demonstrate that cross-modal representations converge in general; it is one measured point, at one model scale, in one language and one image domain. It does establish that linear cross-modal correspondence is measurable far above chance without joint training, and it converts the limitation stated in Section 5 - that this study samples two points of a hypothesised trend and cannot observe its slope - into a concrete next measurement: repeating the identical protocol with a small, base and large image encoder yields three points against

capacity, which is the variable the hypothesis is actually about. Two cautions apply to that sweep: DINOv2's output width grows with model size, so the number of pairs must grow with it to hold rows per input dimension constant, and the widths should additionally be matched by projection so that a rising line reflects convergence rather than fit capacity. A rising line would be direct evidence for scale-dependent convergence; a flat one would be an equally reportable negative.

C.9 Post-submission experiment series (E1.1, F1-F5): caption protocol, distillation control, refit adapter, metric degeneracy

Why five captions, and what averaging changes. COCO supplies five independent human captions per image; five is the dataset's structure, not a tuned choice. The decision was whether to use one or all. A single caption carries one annotator's emphasis - one names the activity, another the setting - so the target mixes image content with individual phrasing. Averaging the five embeddings cancels the idiosyncrasies and retains what they agree on, making the target 'what the image is about' rather than 'what one person wrote'. The effect is large: held-out R-squared rises from 0.278 to 0.511 and R@1 from 0.160 to 0.358. The same averaging must be applied to the ceiling, since it eases the task for any method - SigLIP itself scores 0.648 on single captions and 0.875 on averaged ones. Scoring an averaged system against a single-caption ceiling produced an inflated 55.2 percent; matched properly it is 40.9. Both protocols are reported in Table C6 and both fall in the same pre-registered band, so the verdict does not depend on the choice. The standard COCO retrieval convention is one caption per query, so averaging is a deliberate and disclosed deviation, adopted to lower target noise rather than to raise the score.

The MLP control, retested at scale (F1-F2). Section 4 reads the MLP's +0.003 gain as evidence that the MobileCLIP-to-SigLIP relationship is linear. That reading had an untested alternative: the MLP (1.31M parameters) saw only 3,000 examples and, unlike a linear map, does not decompose into independent per-output regressions - it may simply have been data-starved. A 38,000-image teacher corpus (SigLIP 2 targets; no captions required, since the teacher supplies the target) settles it: retrained at 74 rows per input dimension, the MLP's held-out cosine converges to exactly the linear baseline - 0.9052 against 0.9052, gain -0.0000, plateau by epoch 10. The data-starvation reading is refuted; linearity is confirmed by a second, stronger test. The governing ratio differs by architecture: rows per input dimension for the decomposable linear map, examples per parameter (here 0.029) for the MLP - the plateau, not the ratio, is the evidence.

The shipped adapter improves by refitting alone (F5). The shipped W was fitted at 5.9 rows per input dimension - above this report's threshold of 5 but at the bottom of its 5-10 band. Refitting the identical closed-form ridge on the 38,000 corpus rows and evaluating on 5,000 fresh queries across five disjoint 1,000-image galleries (train2017 images beyond the corpus, unseen by both maps) yields significant gains at every rank under a paired bootstrap:

Metric	shipped W	refit W	paired gain	95% CI	verdict
R@1	0.565	0.590	+0.0244	[+0.0162, +0.0330]	real
R@5	0.847	0.858	+0.0116	[+0.0054, +0.0178]	real
R@10	0.918	0.929	+0.0106	[+0.0060, +0.0152]	real

Table C7: refit-adapter evaluation, pooled over 5,000 queries; 300 gained against 178 lost at rank 1. The artifact is unchanged in kind - the same 0.79 MB matrix, the same seconds-long refit on unlabeled images.

The pre-registered R@1 gate of 90 percent of ceiling, recorded as missed in Section 4.3 at 88.4 percent, is best described as **approached rather than met**: the refit map reads 90.6 percent on the original 1,000-query protocol but 89.0 percent on the expanded evaluation. A smaller-sample point estimate that crossed the line did not survive five times the data - the lesson of Section C.1, applied to evaluation rather than to fitting.

A metric-degeneracy result (E1.1). Replacing the contrastively trained text encoder of Section C.8 with GPT-2 mean-pooled - all else identical, a matched pair - retains 74 percent of the held-out R-squared (0.377) but only 27 percent of R@1 (0.095, still 95x chance; shuffle gap 0.505). The mechanism was measured before fitting: GPT-2's caption space has an effective rank of 6.5 of 768 and a mean inter-caption cosine of +0.999, against DINOv2's 380.2 and +0.031. In that space cosine is degenerate - it reads 1.000 for a map whose retrieval runs at 9.5 percent - the mirror image of Section 4, where R-squared understated a system whose cosine was honest. Together: no similarity metric is universally correct; the geometry of the space determines which metrics are interpretable, and it must be measured first. A label-free rescue test (centering, top-component removal, whitening; notebook E1.2) is built and pending - it separates

'correspondence absent' from 'isotropy absent'.

An initial scale-sweep point (E1, dinov2-large). Section C.8 notes that this study samples a hypothesised trend at one encoder size and cannot observe its slope. A first additional point has been measured: DINOv2-large with cls+patch pooling - the CLS token concatenated with the mean of the patch tokens, DINOv2's own linear-probe protocol, giving a 2,048-dimensional input - fitted on 22,755 rows (11.1 per input dimension) against the same averaged-caption targets. It is the strongest cross-modal result in the project: held-out R-squared 0.609, above the same-modality vision pair's 0.592; cosine 0.912; R@1 0.504, or 58.3 percent of a matched ceiling of 0.865, against 40.9 percent for the base encoder; shuffle gap 0.709; raw floor at chance. R@10 reaches 94.2 percent of ceiling.

This is not yet a capacity measurement. Four factors differed between the two runs - encoder (base to large), pooling (cls to cls+patch), input width (768 to 2,048) and sample count (8,000 to 23,755) - so the 17.4-point gain cannot be attributed to encoder capacity. Sample count is a deliberate control, scaled to hold rows per input dimension near 10, and width is largely neutralised by that same scaling, since rows per parameter then stays constant. Pooling is the substantive confound: cls+patch changes what information the vector carries, not merely how many dimensions hold it, and is known to improve linear probes independently of model size. A matched slope therefore requires the smaller encoders re-run under cls+patch, which is in progress; MATCH_DIM offers an additional capacity-controlled variant.

Note on regularization. ALPHA_GRID selects ridge regularization strength on a validation slice carved from the training rows; it is a search, not a learning-rate schedule, since the ridge solution is closed-form. Its sensitivity is itself diagnostic: in the dinov2-large run every candidate scored 0.596, the problem being well-conditioned at 22,755 rows against 2,048 dimensions, whereas in E1.1 an alpha of 10 produced R-squared -0.741 in GPT-2's near-degenerate space.

Training status across the project. Experiments A, B and E fit only closed-form matrices under no_grad; every pretrained model remains bit-identical to its download. The F series is the deliberate contrast condition: F2 trains a separate head beside the frozen encoders; F3, the only notebook that would modify encoder weights, is built and deliberately unrun.

C.9.1 The linearity result, reproduced twice from a fresh corpus

Finding 1 - that a non-linear head converges to exactly the linear answer - was re-run end to end after all other work was complete: the distillation corpus rebuilt from scratch (40,000 images, 37,999 training rows at 74.2 rows per input dimension), the teacher checkpoint verified by a transfer check against the project's own artifacts (cosine 0.9168, the same value recorded when the corpus was first built), and the head distilled twice under identical settings, differing only in training stochasticity.

quantity	run 1	run 2	what it shows
held-out cosine, linear	0.9183	0.9183	the closed-form baseline on this corpus
held-out cosine, MLP	0.9183	0.9183	identical to four decimals, both runs
capacity gain (cosine)	+0.0000	+0.0000	the CONTROLLED comparison: same corpus, same protocol, only capacity differs
R@1 vs the SHIPPED adapter	+0.0320	+0.0270	confounded - moves data AND capacity together
residual after F5's data effect (+0.0244)	+0.0076	+0.0026	varies by more than half its own magnitude between runs

Table C7b: two independent runs of the same notebook on the same corpus.

The stability is the point. The controlled quantity - linear against MLP on the same rows under the same protocol - is identical to four decimal places in both runs. The confounded quantity moved by two thirds of its own size with nothing changed but initialisation and batch order. A residual that varies more between runs than it measures is not a small real effect; it is noise, and observing it wobble is better evidence of that than the standard-error argument alone.

Why the R@1 row is reported at all. It compares the distilled head against the SHIPPED adapter, which was fitted on 3,000 pairs, so it differs in training data as well as in capacity. Section C.9 measured the data effect in isolation - the same closed-form ridge refitted on 38,000 rows - at R@1 +0.0244. Subtracting it leaves the residuals above. The row is kept because removing it would hide the comparison a reader would naturally make; it is annotated because leaving it unqualified would let a data effect be read as a capacity effect.

What this does for Finding 1. The claim moves from a single measurement to a reproduced one. The absolute cosine differs from the figure quoted in Section C.9 (0.9052) because the corpus is a different draw; the EQUALITY, not the level, is the finding, and it survived a complete rebuild of the corpus and two independent training runs.

C.10 The scale sweep: cross-modal alignment against image-encoder capacity

Section 5 lists as a limitation that this study samples two points of a hypothesised trend and cannot observe its slope. The Platonic Representation Hypothesis is fundamentally a claim about that slope - that representational alignment grows with model capability - so a two-point sample is consistent with it but cannot test it. This section replaces that limitation with a measurement.

Design. The cross-modal protocol of Section C.8 was repeated with three image encoders of the same family and training procedure - DINOv2 small, base and large, 22M to 304M parameters, a 13.8-fold range - holding every other factor fixed: the same text encoder (bge-m3), the same COCO images and five-caption averaging, the same cls-plus-patch pooling, the same ridge protocol with a validation alpha sweep, and the same direction- and treatment-matched SigLIP ceiling measured per run. Two controls make the comparison a capacity measurement rather than a confound. First, pooling is held constant: an earlier base run used CLS-only pooling and is excluded, because concatenating the patch mean changes what information the vector carries and is known to help linear probes independently of model size. Second, the number of pairs scales with the encoder's output width so that rows per input dimension is 11.1 at every point - without this, wider encoders would be progressively starved, and starvation biases downward, which would understate exactly the trend under test.

Encoder	Para ms	Width	Pairs	rows/di m	R2	shuffle gap	R@1	R@5	R@10	% ceiling @1 / @5 / @10
DINOv2-small	22M	768	9,533	11.1	0.516	0.616	0.357	0.698	0.826	40.9 / 71.3 / 83.4
DINOv2-base	87M	1,536	18,066	11.1	0.580	0.682	0.466	0.809	0.907	53.5 / 82.2 / 91.2
DINOv2-large	304M	2,048	23,755	11.1	0.609	0.709	0.504	0.835	0.935	58.3 / 85.6 / 94.2

Table C8: the sweep. Width is the pooled output (CLS concatenated with the patch mean, hence twice the hidden size). Ceilings were measured per run on that run's own held-out rows and fell between 0.865 and 0.872 at R@1, so the percentage column is comparable across rows.

Result. Every measure rises monotonically with capacity: held-out R-squared, mean cosine, all three recall levels, and the shuffle-control gap. Six measures moving together across a 13.8-fold parameter range is a considerably stronger signal than any one of them alone, and the rising shuffle gap is worth noting separately - it shows the fit is pulling more signal relative to its own misaligned control, not merely scoring higher. Against the logarithm of parameter count the trend is close to linear: R@1 gains 15.3 points of ceiling per decade (fit R-squared 0.949) and held-out R-squared gains 0.082 per decade (0.966). Both base and large exceed this study's same-modality reference of 0.592 from Section 4, on a task that crosses modalities without joint training.

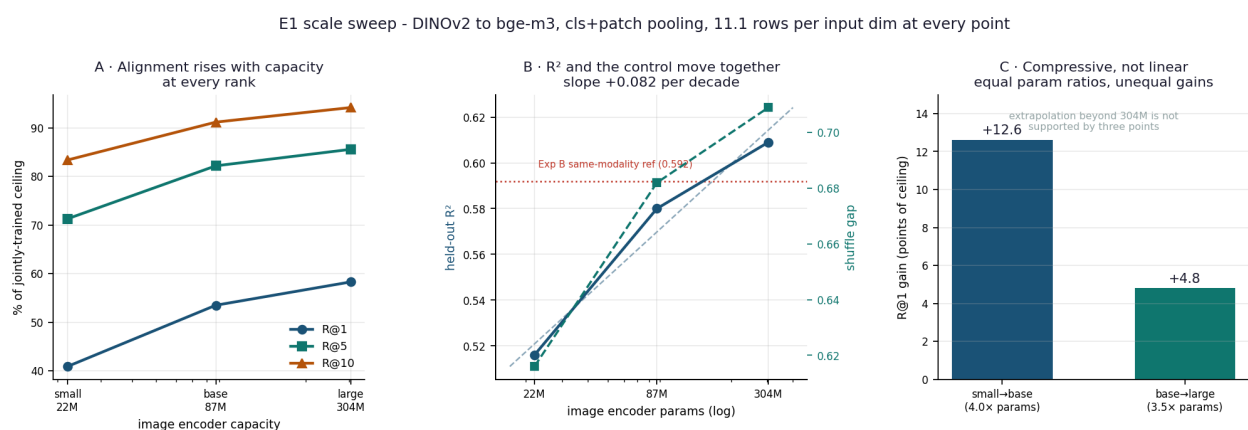


Figure C4: alignment against capacity. Panel A, share of a jointly-trained ceiling at each rank. Panel B, held-out R-squared with the shuffle control on the secondary axis, both against log parameters. Panel C, the gains from equal parameter ratios.

The returns are compressive. Roughly equal parameter ratios do not buy equal improvement: small to base is a 4.0-fold increase and gains 12.6 points of R@1 ceiling share, while base to large is 3.5-fold and gains 4.8. The log-linear fit is therefore a convenient summary of three points rather than a law, and this section makes no claim about behaviour beyond 304M parameters. A flattening curve is what one would expect as the ceiling is approached - large already reaches 94.2 percent of it at rank ten - but three points cannot distinguish

saturation from a genuinely diminishing exponent.

What this establishes. Within the range measured, representational alignment between a vision encoder that has received no language supervision and a text encoder that has never seen an image increases with the vision encoder's capacity. That is the Platonic Representation Hypothesis's central prediction, tested in the setting where it is least trivially true, with pooling and statistical power held constant across points. It remains one model family, one text encoder, one image domain and one language, and the appropriate reading is a measured trend within those bounds rather than a general law. The limitation stated in Section 5 - that the slope could not be observed - is discharged; the narrower limitation that replaces it is that the slope has been observed over 13.8-fold, in one family, and is compressive at the top of that range.

C.11 Isotropy, not similarity training: a label-free correction recovers the collapsed retrieval

Section C.9 reported that substituting GPT-2 mean-pooled for the contrastively trained text encoder retains most of the held-out R-squared but collapses retrieval, and concluded that usable retrieval geometry requires similarity training. That conclusion conflated two claims it never separated: that the correspondence is absent from raw language-model representations, and that the coordinate frame merely hides it. This section separates them, and the second is correct.

Design. Four corrections were applied to GPT-2's caption space and the identical fit-and-evaluate protocol re-run under each. All are purely geometric and label-free - no contrastive objective, no pairs, no gradient - and every transform is estimated on the training targets only and applied unchanged to the held-out rows. Image side: DINOv2-base with cls-plus-patch pooling, 17,066 training rows at 11.1 per input dimension.

Correction	eff. rank (of 768)	R2	R@1	% of matched bge-m3 arm
DINOv2-base (bge-m3 arm: R@1 0.466)				
raw (C.9 baseline)	6.5	0.440	0.130	27.9%
centered	6.5	0.440	0.104	22.3%
all-but-top-1	18.1	0.402	0.156	33.5%
all-but-top-3	34.7	0.464	0.227	48.7%
all-but-top-10	99.8	0.365	0.320	68.7%
whitened	490.8	0.014	0.478	102.6%
DINOv2-large (bge-m3 arm: R@1 0.504)				
raw	6.4	0.439	0.136	27.0%
centered	6.4	0.439	0.131	26.0%
all-but-top-1	18.3	0.416	0.164	32.5%
all-but-top-3	35.2	0.465	0.252	50.0%
all-but-top-10	101.8	0.386	0.374	74.2%
whitened	495.2	0.034	0.511	101.4%

Table C9: label-free corrections on GPT-2's caption space, at two image-encoder scales. Percentages are against the bge-m3 arm measured with the SAME encoder and pooling - drafts that compared against a different configuration overstated the recovery in both cases, which is why the notebook now derives the reference arm from the pairs filename rather than accepting it as a setting.

Result, replicated at two scales. Whitening the target space - a linear, unsupervised rescaling estimated from training targets alone - lifts R@1 from 0.130 to 0.478 with DINOv2-base and from 0.136 to 0.511 with DINOv2-large: lift factors of 3.7 and 3.8, reaching 102.6 and 101.4 percent of an encoder contrastively trained on 1.2 billion text pairs for exactly this purpose. The rank correlation between effective rank and R@1 is +0.94 at both scales. The correspondence was present in the raw language-model representations throughout; only the coordinate frame concealed it. The claim that similarity training is required does not survive: what similarity training supplies here is isotropy, and isotropy can be supplied afterwards, for free.

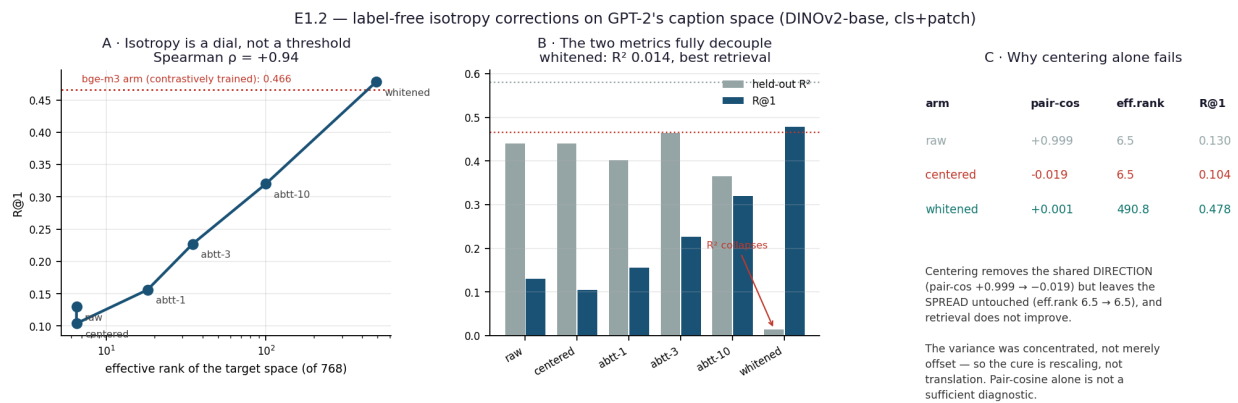


Figure C5: isotropy against retrieval (A), the divergence of R-squared and recall (B), and why centering alone is insufficient (C).

Isotropy behaves as a continuous dial. Across the six arms, effective rank rises 6.5, 6.5, 18.1, 34.7, 99.8, 490.8 and R@1 rises 0.130, 0.104, 0.156, 0.227, 0.320, 0.478 - a rank correlation of +0.94. Retrieval quality is not gated by crossing an isotropy threshold; it tracks the degree of isotropy monotonically.

Centering alone fails, and the failure is instructive. Centering drives mean pair-cosine from +0.999 to -0.019 - the standard symptom of anisotropy, apparently cured - while leaving effective rank unchanged at 6.5, and retrieval does not improve (0.130 to 0.104). The variance was concentrated across directions, not merely displaced by a common offset, so the remedy had to be rescaling rather than translation. Mean pair-cosine is therefore not a sufficient diagnostic on its own; it must be read alongside a spectral measure.

The metrics separate completely. The whitened arm records a held-out R-squared of 0.014 - indistinguishable from zero - while delivering the best retrieval of any arm and beating the contrastively trained reference. Whitening rescales the target space, so predicting it in squared-error terms becomes nearly impossible, while the angular structure that retrieval consumes becomes nearly ideal. Taken with Section 4, where R-squared of 0.592 understated a system whose cosine was 0.900, and Section C.9, where cosine of 1.000 overstated a system retrieving at 9.5 percent, this completes the argument: a similarity metric is interpretable only relative to the geometry of the space it is computed in, and that geometry must be measured first.

This resolves an apparent contradiction with Section C.10. Measured raw, the GPT-2 arm stops responding to image capacity: moving from DINOv2-base to DINOv2-large gains 0.002 in R-squared and nothing in R@1 (0.160 to 0.138, a difference of 22 queries in 1,000 and within noise), while the bge-m3 arm over the same two encoders gains cleanly at every rank. That looked like the extra capacity being wasted against a degenerate target. It was not: after whitening, the large encoder does benefit - 0.478 to 0.511 - tracking the bge-m3 arm's own base-to-large gain. The additional information was arriving throughout; the coordinate frame was discarding it. The scaling trend of Section C.10 therefore holds against a language-model target as well, but is invisible until the frame is corrected - which is itself the strongest form of this section's argument.

Implication beyond this project. Cross-model alignment is routinely quantified with cosine- and kernel-based metrics. Where either space is strongly anisotropic, those metrics can report an absence of alignment that a three-line unsupervised transform removes. Reported failures of convergence should therefore be checked against an isotropy battery before being believed - the measurement may be describing the coordinate frame rather than the content.

C.11.1 The collapse itself, drawn

Three of this appendix's findings turn on a single phenomenon, so it is worth isolating. GPT-2's mean-pooled caption vectors have an effective rank of 6.5 out of 768 and a mean pairwise cosine of +0.999: one direction carries 59.6 percent of all variance, and every caption sits in a cone a couple of degrees wide. Cosine between any two of them reads near 1.000 regardless of what they say.

The origin is the training objective. GPT-2 optimises next-token prediction and nothing else; it has no CLS token, because its attention is causal and a prepended summary token would see nothing, no pooling layer, and no similarity loss. It never produced a sentence-level vector during training - that vector is this project's construction, a mean over token states optimised for a different task - and averaging surfaces the large component every passage shares.

The consequences are the three results already reported. The fitted map scores cosine 1.000 while retrieving at R@1 0.160, with held-out R-squared 0.414: the content IS predicted, at 71 percent of the bge arm, and cosine simply cannot read it. A label-free linear transform corrects it, raising effective rank to 490.8 and R@1 to 0.478. And it forms the bottom rung of the pre-registered ladder in Section C.14, where raw retrieval across four training objectives ordered by effective rank with rank correlation +1.00.

The GPT-2 collapse: a space that encodes the content and hides it from cosine

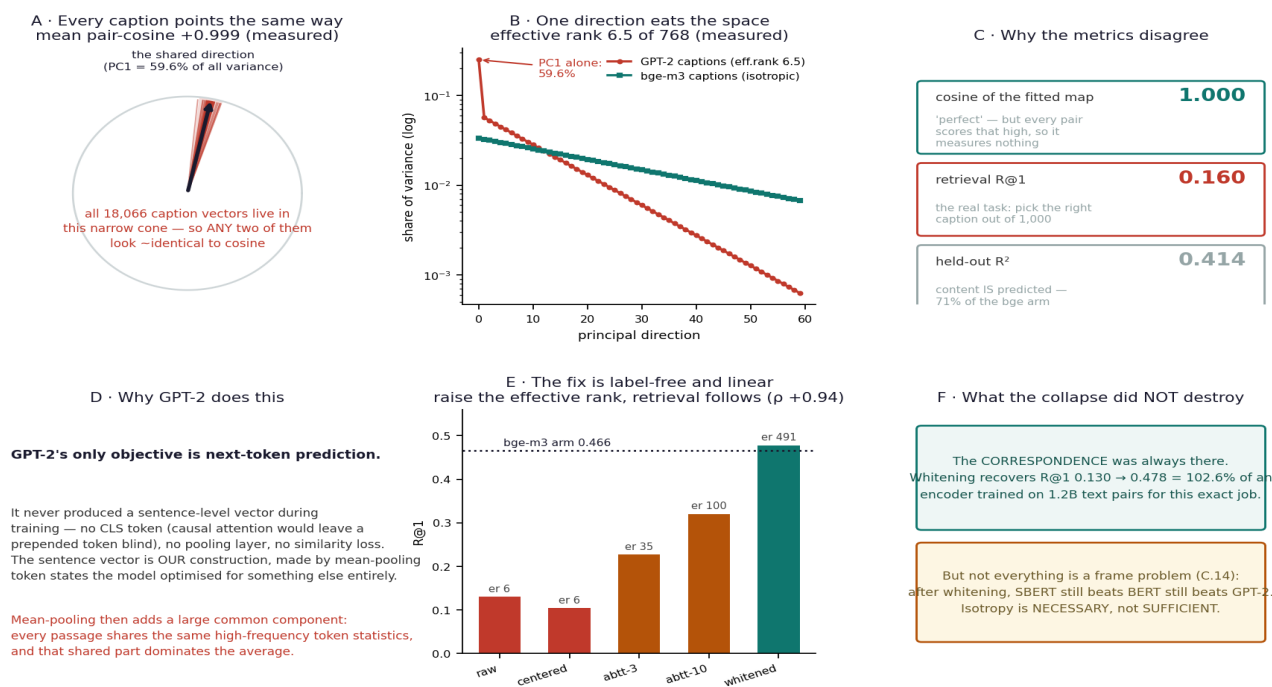


Figure C5b: the collapse and its spectrum, the metric disagreement it causes, its origin, the label-free fix, and what it does not destroy. The point cloud in panel A is schematic; every number shown is measured.

C.11.2 Shape before coordinates: what agrees without any fitted map

Every measurement so far compares two spaces THROUGH a map fitted between them, which invites an obvious objection: the agreement might be manufactured by the fit. This section removes the map entirely. Each space is compared only to ITSELF - how far apart are items i and j inside the phone space, and separately inside the server space - and the two answers are then correlated. Nothing is fitted, nothing is aligned, and the measure is invariant to orientation, scale and even dimension.

measure	value	chance / reference	what it shows
Spearman correlation of pairwise cosine distances	+0.681	0 if unrelated	items that are far apart in one space are far apart in the other
Pearson, same quantities	+0.763	0 if unrelated	the relation is close to monotone and roughly linear
10-nearest-neighbour overlap	67.1%	2.5% (27x)	two thirds of an item's closest neighbours are the same items in both spaces
cross-space cosine, raw	~0.00	-	the SHAPES agree while the COORDINATES do not
mean cosine to own twin, after the 0.79 MB map	0.902	-	one linear map supplies the missing frame

Table C10b: shape agreement measured on 400 held-out images, with no fitted map in the first three rows.

Geometry similarity: the same shape in two unrelated coordinate frames

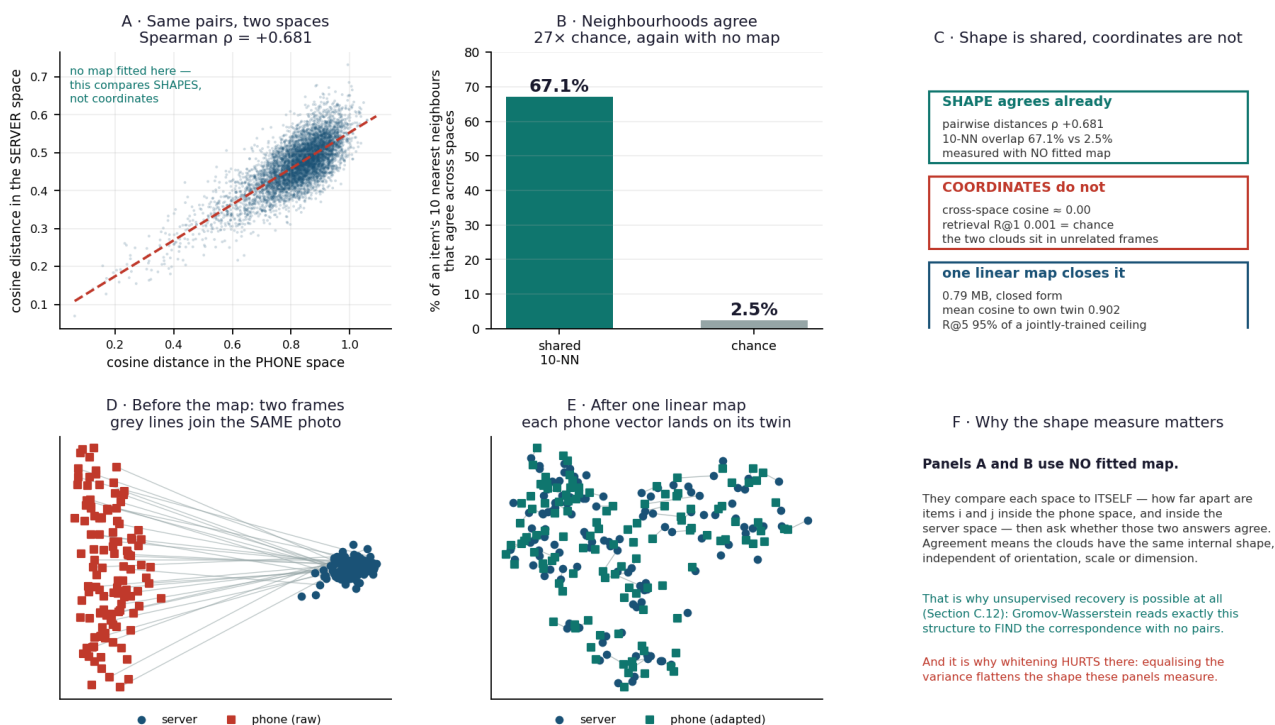


Figure C5c: pairwise distances agree (A) and neighbourhoods agree (B) before any map exists; the two clouds nonetheless occupy unrelated frames (D) until one linear map lands each phone vector on its twin (E).

Why this strengthens the thesis rather than restating it. The convergence claim can be read two ways: that a fitted map can be made to work, or that the structure was already shared and only the coordinate frame differed. These numbers separate them. Two thirds of nearest neighbours already agree at 27 times chance while cross-space cosine sits at zero - so the agreement cannot have been created by a fit that has not happened yet. The adapter is not building a correspondence; it is supplying coordinates for one that is already there.

This is also the quantity Section C.12 exploits. Gromov-Wasserstein matches two point clouds using nothing but each one's internal distance matrix - exactly the structure measured in panels A and B - which is why unsupervised recovery is possible at all. And it explains the negative result there: whitening equalises variance

across directions, which flattens precisely the shape these panels measure, and unsupervised matching duly collapsed from 39.7 percent to 0.1. The same measurement that explains why discovery works explains why isotropy breaks it.

C.12 Recovering the correspondence without any paired examples

Every experiment to this point is given the pairing: row i of one array is the same item as row i of the other, and only the map is fitted. This section removes that last piece of supervision. Two sets of 1,500 vectors are taken from the same images, one side is shuffled, the true permutation is sealed away and used only for final scoring, and the correspondence itself must be recovered from the shapes of the two point clouds. Stage one matches structure to structure with entropic Gromov-Wasserstein - each space is described only by its own internal cosine-distance matrix, so no cross-space pair is ever seen. Stage two fits a ridge map from the assignment that stage one proposes.

Result. Exact matching reaches **39.7 percent** against a chance rate of 0.067 percent - 595 times chance - with 53.5 percent of true partners in the top five and 58.0 in the top ten. The map fitted from that assignment reaches R@1 0.252 and R@10 0.567 in the standard retrieval protocol, which is 40 percent of the jointly-trained ceiling and 45 percent of the supervised adapter, achieved with no paired examples whatsoever.

Map	paired examples used	R@1	R@5	R@10	verification AUC
ceiling: SigLIP native	-	0.630	0.885	0.944	1.000
supervised W	3,000	0.557	0.842	0.918	0.999
unsupervised W	0	0.252	0.477	0.567	0.906

Table C10: the unsupervised map against the supervised one and the ceiling, on the same held-out rows. Verification AUC asks whether a threshold separates a true pair from a random one - a weaker question than outranking 999 impostors, and the unsupervised map answers it far better than its retrieval figures suggest.

The entropic regularizer dominates the outcome. An earlier run of this experiment reached only 8.6 percent exact matching and produced a map at 6 percent of ceiling. The difference was one hyperparameter: that run used $\epsilon = 5e-4$ with 200 iterations, under which Sinkhorn did not converge. Too small a regularizer makes each projection nearly a hard assignment, which both converges slowly and locks in early mistakes.

epsilon	exact	top-5	top-10	x chance	seconds
1e-3	39.7%	53.5%	58.0%	595x	273
3e-3	12.2%	26.5%	32.6%	183x	28
5e-3	6.6%	16.2%	22.5%	99x	18
1e-2	1.3%	4.6%	8.1%	19x	15

Table C11: the sweep at $N = 1,500$. The optimum is sharp and it moves with N - a preliminary sweep at $N = 800$ placed the best value at $5e-3$, which at $N = 1,500$ yields 6.6 percent. The regularizer must be tuned together with the problem size rather than carried across settings, which is why the notebook sweeps rather than fixes it.

A deliberate negative: whitening destroys the recovery. Section C.11 established that whitening the target space rescues cosine retrieval, lifting R@1 by a factor of 3.7. Applying the same transform to both spaces before Gromov-Wasserstein collapses matching from 39.7 percent to 0.1 percent, against a chance rate of 0.067 - complete destruction of the signal, confirmed at two problem sizes. The effective ranks confirm the transform did what it claims: 135 and 171 before, 494 and 743 after.

This is the sharpest distinction the project produces, and it is not a contradiction. Cosine retrieval reads a correspondence that is already known; there, variance concentrated in a few directions is noise that drowns the distinctions retrieval needs, and equalising it helps. Gromov-Wasserstein discovers a correspondence that is unknown, by matching one space's distance structure against the other's; there, uneven variance is precisely the signal, because it is what makes a point's position distinctive relative to its neighbours. Whitening equalises every direction and so erases exactly the structure the matcher depends on. Isotropy helps when reading a correspondence and hurts when finding one - the same geometric property, opposite prescriptions, both measured rather than assumed.

A second negative: iterative refinement degraded the solution. Alternating between fitting a map and re-matching - the standard ICP refinement - reduced accuracy monotonically across ten iterations (8.6, 8.3, 7.9, and so on to 6.9 percent) and converged stably at the degraded answer. Fitting a map from mostly incorrect

assignments teaches the incorrect correspondence, which is then re-matched onto with greater confidence; convergence there indicates agreement with its own errors rather than correctness. The result reported above is Gromov-Wasserstein alone.

What this establishes. The correspondence between two independently trained encoders is partially self-identifying: it can be recovered from geometry alone, at 595 times chance, without a single paired example. That is the most demanding form of this project's thesis - not merely that the spaces are related by a linear map, but that the relation is discoverable without being told. It remains far weaker than the supervised map and is evidence about the geometry rather than a deployable component; the honest summary is substantial partial recovery, with the mechanism of both its success and its two failures measured.

C.12.1 Scope: which encoder pairs carry which claim

A convergence result is only as strong as the independence of the two models behind it, and that independence is not uniform across this project. Two encoders trained on the same kind of objective might agree because both approximate the same task, rather than because independently trained networks converge on a shared representation of the world. The distinction is worth stating explicitly, because the experiments differ on it and the strongest claim does not come from the strongest-looking number.

experiment	encoder pair	shared training paradigm?	strength against the 'same task' objection
C.12 (unsupervised recovery)	MobileCLIP-S1 and SigLIP 2	YES - both contrastive image-text models, though from different organisations, on different data, with different losses (sigmoid vs softmax)	weakest. Both were trained to place images near their captions, so some shared geometry is expected before any convergence claim is made
Section 3 (Experiment A)	GPT-2 and Pythia-160M	NO similarity objective on either side; different tokenizers, positional schemes, corpora and organisations	stronger. Neither model was ever optimised to make similar inputs produce similar vectors
C.8-C.11, C.14 (cross-modal)	DINOv2 and bge-m3 / BERT / SBERT / GPT-2	NO - DINOv2 received no language supervision at any point; the text encoders never saw an image	strongest. No objective on either side rewards agreement with the other modality, and the effect nonetheless scales with capacity (C.10)

Table C11b: the independence of each pair, and what it costs or buys the claim built on it.

What this means for Section C.12. The measured result is unaffected: 39.7 percent exact matching at 595 times chance, with the true permutation sealed until scoring, is what it is. What narrows is the claim it supports. The honest statement is that the correspondence between two CONTRASTIVELY TRAINED IMAGE ENCODERS is partially recoverable from geometry alone - not that this holds between arbitrary models. Whether unsupervised recovery survives on a pair with no shared objective is untested here, and is the obvious next experiment: the shape-agreement diagnostic added to G1 and G3 measures exactly the precondition it depends on, and can be run on any pair before attempting the matching.

And what it means for the use of bge-m3 as a target. bge-m3 is contrastively trained, which invites the same question in reverse: is a result that reads out into a similarity-optimised space circular? It is not. bge's contrastive training operated on text-text pairs; it never saw an image, and DINOv2 never saw text, so neither objective rewards agreement with the other. What bge does supply is a coordinate frame in which cosine is meaningful - and rather than assuming that contribution, Sections C.11 and C.14 measure it: whitening a raw language-model space recovers 102.6 percent of the bge arm, and the residual 0.110 spread across the ladder is what contrastive training adds beyond the frame. The target's advantage is quantified, not presupposed.

C.13 The shared space as an object: one hub, five encoders, zero-shot component transfer

Every preceding result is pairwise: encoder A relates to encoder B through a map fitted for that pair. N encoders would need N-squared maps, and nothing in a collection of bilateral fits shows that the encoders share ONE coordinate system. This section constructs that coordinate system and tests it by use. Five cached spaces over the same 9,533 images - DINOv2 small, base and large (no language supervision), bge-m3 captions (contrastively trained), and GPT-2 captions (next-token prediction only) - are each given a single linear map into a 512-dimensional hub: a whitened PCA basis of all five spaces concatenated after per-space standardisation, estimated on 8,533 training rows. Row alignment across caches is verified numerically before anything is fitted, and 1,000 rows are held out for every measurement.

The constructive test. A caption-retrieval head (hub to bge-m3 space) is trained on the hub vectors of ONE encoder - DINOv2-small - and then applied, unchanged, to the hub vectors of encoders it has never seen in any form. A random map into the hub replaces the fitted one as a control, and must sit at chance for the test to mean anything.

encoder	R@1	R@5	R@10	% of native head	
img_small	0.398	0.739	0.875	111.5%	trained here
img_base	0.442	0.792	0.902	95.9%	never seen
img_large	0.468	0.808	0.914	92.9%	never seen
random-map control	0.000	-	-	-	chance = 0.001

Table C12: zero-shot transfer of a single trained component. 'Native head' is a head fitted directly on that encoder's own hub vectors - the upper reference for each row.

Result. The transferred head reaches 95.9 and 92.9 percent of the native heads on two encoders it never saw. Absolute performance RISES on the unseen encoders - 0.398, 0.442, 0.468 - so the component is not merely surviving transfer: a stronger encoder still helps, through the shared coordinates, using a head fitted on a weaker one. The control sits at chance. A working system has been assembled from parts that were never trained together, which is the Platonic Representation Hypothesis's constructive core: the shared representation exhibited as an object and used, rather than inferred from correlations.

The operating point was measured, not chosen. Hub width and ridge strength were swept jointly, with the project's minimum of five training rows per fitted dimension enforced:

HUB_DIM	rows/dim	a=0.01	a=0.1	a=1.0	a=10
128	66.7	0.390	0.390	0.390	0.390
256	33.3	0.430	0.430	0.430	0.428
512	16.7	0.442	0.442	0.445	0.445
768	11.1	0.006	0.013	0.092	0.341
1024	8.3	0.007	0.009	0.024	0.137

Table C13: transfer R@1 (img_small-trained head on unseen img_base) across hub width and ridge alpha. Verification AUC at the chosen point is 0.979-0.984.

There is a cliff between 512 and 768 dimensions, and it is not a statistical-power artifact - rows per dimension remains above the threshold on both sides. The hub basis is a whitened PCA, which divides each component by its singular value; past the cliff the basis amplifies low-variance directions whose content is noise, and those directions then dominate the cosine geometry that transfer depends on. Below the cliff, alpha is irrelevant across three orders of magnitude; past it, heavy regularisation claws back part of the loss (0.006 to 0.341 at 768) but never reaches the narrow hub's level. Width selection matters; regularisation cannot substitute for it.

Transitivity, and an honest trade-off. Composition through the hub (A to hub to B, with A-to-B never fitted) retains 83 to 95 percent of directly-fitted maps for every image-space target - essentially meeting the pre-registered 85 percent bar - and fails into the two text spaces, whose reconstruction the 512-d hub does not

carry (their reach-into-hub R-squared, 0.24 and 0.19, predicts exactly this). Widening the hub repairs reconstruction - retention into the widest targets reached 84 to 88 percent at 2048 in a separate measurement - but that same widening is what destroys transfer. The two demands pull the width in opposite directions, and no setting satisfies both. A hub tuned for portability is not a lossless reconstruction medium, and vice versa: this is the reading-versus-representing tension of Sections C.11 and C.12 in a third form, and it appears to be a property of shared linear coordinates as such, not of any one experiment.

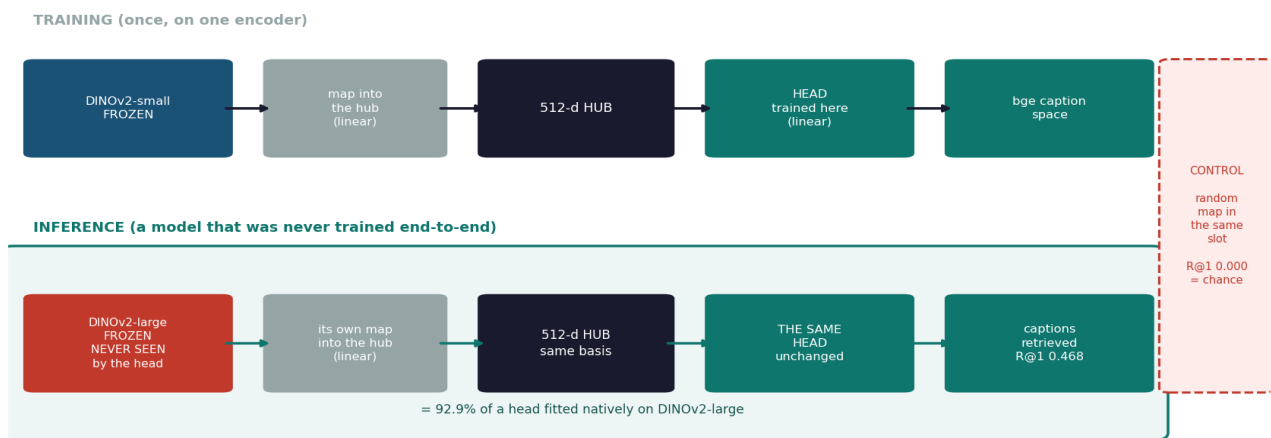
Scope. Five spaces from two model families, one image domain, one language. What is exhibited is a shared linear coordinate system for these five encoders, demonstrated by component transfer at 93 to 96 percent of native performance - not a universal space for all models. Within that scope, the central objection to convergence claims - that pairwise fits might each be learning their own idiosyncratic bridge - is answered: one basis serves all five, and work done in it moves between encoders that never met.

C.13.1 What the hub assembles, drawn

The transfer result is easier to misread as a fitting exercise than to read correctly, so it is worth drawing. In training, one encoder's frozen output is mapped into the hub and a single linear head is fitted from hub coordinates to the caption space. At inference the encoder is REPLACED by one the head has never seen, which supplies its own linear map into the same hub; the head is not retrained, refitted, or adjusted. The resulting pipeline - frozen encoder, linear map, unchanged head - was never trained end to end, and it works at 92.9 percent of a head fitted natively on that encoder.

Two properties make this a statement about representation rather than about regression. The control substitutes a random matrix in the map slot alone and collapses to chance, so nothing is being carried by the head by itself. And absolute performance RISES with the substituted encoder's quality (R@1 0.398, 0.442, 0.468), which a head that had merely memorised the task could not produce. Swapping the encoder requires no retraining anywhere: that portability is what a shared coordinate system buys.

What the hub actually builds: a NEW model assembled from parts that never trained together



The only trained-from-scratch part is one linear head. Every encoder is frozen and pretrained; every map is a closed-form ridge solve.

Swapping the encoder does not require retraining anything — that is what “a shared coordinate system” buys, and what makes the assembly a new model rather than a new fit.

Figure C6: training once on one encoder, then running inference through a different one. Every box except the head is frozen or a closed-form solve.

C.13.2 Lineage: cross-family head transfer, now measured

The transfer result of Section C.13 was first obtained within the DINOv2 family. DINOv2-S, -B and -L are distilled from a common ViT-g/14 teacher: they never trained jointly, but they descend from shared weights on the same data, so transfer between them demonstrates consistency across model SCALE within one family, not across independent lineages. Experiment G4 closes that gap directly.

The test. A caption head was trained on DINOv2-small's hub coordinates alone, then applied UNCHANGED to SigLIP 2 - a vision encoder from a different organisation, teacher, training corpus and objective, with no DINOv2 ancestry. SigLIP 2 was encoded over the same 9,533-image prefix, given one linear map into the FROZEN hub, and read out through the existing head.

recipient encoder	lineage vs the head	R@1	% of a natively-fitted head
img_base	ViT-g student (sibling)	0.445	96.5%
img_large	ViT-g student (sibling)	0.473	93.8%
SigLIP 2	INDEPENDENT - no DINOv2 ancestry	0.557	94.2%
random-map control	-	0.003	chance (0.001)

Table C12g: cross-lineage transfer lands inside the within-family band.

The result. SigLIP 2 retains 94.2 percent of a head fitted natively on it - inside the 93.8 to 96.5 percent measured for the two DINOv2 siblings. Shared ancestry was therefore NOT what made the within-family transfer work: the head reaches an unrelated lineage as well as it reaches a sibling. The control, a random map in the same slot, collapses to chance, so the hub map rather than the head is carrying the transfer; and absolute R@1 rises with the recipient's own quality (0.445, 0.473, 0.557), which a head that had merely memorised the task could not produce.

What is now established, and what is still open. The hub carries components across encoders that never met, and G4 shows this survives a change of LINEAGE. Two boundaries remain, and both are worth stating plainly. First, SigLIP 2 is contrastively trained on image-text pairs, so while its LINEAGE is independent of DINOv2, its OBJECTIVE is well-suited to a caption read-out - the 94.2 percent is cross-lineage, not simultaneously cross-lineage and cross-objective. Second, SigLIP 2 is still a Vision Transformer. The stronger remaining test was a CONVOLUTIONAL encoder, and it has since been run (Appendix E.2): ConvNeXt-base reaches 87.5 percent of native against SigLIP's 86.4 percent measured on the same reconstructed head. Architecture does not bound the claim. The objective confound survives, since ConvNeXt is supervised where DINOv2 is not.

A note on Section C.10. The same shared lineage that the scaling sweep relied on is what made that sweep clean: identical teacher, data and recipe, only student capacity varying, is the control that isolates capacity. The bound stated there still holds - alignment rises with capacity among distilled students of one teacher over a 13.8-fold range - and G4 now adds that the resulting shared space is not confined to that family.

One more control on this protocol, since it bears on the number just quoted. The four-space hub used for Table C12g contains bge-m3, which is also the head's fitting target - an untested dependency until measured. G10 tests it directly: a hub built without bge-m3 (images only) transfers at 85.5 percent of native rather than 95.3, a gap of 9.8 percentage points. Roughly two thirds of that gap (6.8pt) is the generic benefit of any text space sitting in the hub - a matched non-bge control shows a comparable rise for a different target. The remaining 3.0 percentage points is specifically the target sitting inside its own hub's construction. Held-out scoring rules out train/test leakage; the dependency is real and is declared here rather than left implicit (E.6, E.9). Nothing about the 95.3 percent changes: it is still the correct figure for the published protocol, held-out and unchanged.

C.13.3 Seven encoders, four objectives: what one hub carries, and in which direction

Section C.13 built a hub over five spaces. Adding the two encoders from the pre-registered ladder of Section C.14 - BERT and Sentence-BERT - brings it to seven spaces spanning FOUR text training objectives (causal LM, masked LM, MLM-plus-contrastive, heavy contrastive) and two modalities, over the same 9,533 images. The hub is unchanged in construction: a 512-dimensional whitened PCA of all spaces concatenated after per-space standardisation, one linear map per encoder, fitted on 8,533 training rows.

What the extra encoders buy is scope, not performance. The width-by-alpha sweep gives its best transfer at HUB_DIM 512 with alpha 0.01: R@1 0.451 for a head trained on img_small and applied to the unseen img_base, with verification AUC 0.986. The five-space hub gave 0.445 at the same width. That difference is six queries in a thousand - 0.27 standard errors, with a 95 percent interval of -0.038 to +0.050 - and is NOT distinguishable from sampling noise. The correct statement is that transfer is UNCHANGED by adding two encoders, which is itself the useful result: the hub absorbs two further training objectives at no measurable cost to the components already using it. Alpha is likewise irrelevant below the cliff, moving transfer by 0.003 across three orders of magnitude, and the cliff between 512 and 768 dimensions persists at the wider concatenation while remaining above the five-rows-per-dimension floor on both sides - so it is not a power effect.

Composition through the hub, measured scale-free. R-squared cannot be used for this comparison: a target whose vectors are nearly identical has almost no variance, so the denominator of R-squared collapses and any error produces a large negative number. That is this project's own Section C.11 lesson applied to its own diagnostic. R@1 is scale-free and is the verdict metric here.

pairs	mean R@1 retention through the hub
all 41 ordered pairs	87.0%
excluding the 5 pairs writing INTO GPT-2	95.6% (36 pairs, range 68.2-112.0%)
the 5 pairs writing into GPT-2	24.8%

Table C12c: composition through the hub relative to a directly-fitted map for the same pair. Of 41 pairs, six fall below 70 percent and five of those are the single problematic target.

The asymmetry, and its measured cause. Retention depends on DIRECTION, and the dependence tracks the target's geometry:

space	effective rank	as SOURCE (read out of)	as TARGET (written into)
bge-m3	97.4	98.1%	98.1%
BERT	65.0	84.8%	76.6%
GPT-2	6.6	95.1%	24.8%

Table C12d: a collapsed space can be read out of but not written into.

The obvious explanation - that the hub discards the collapsed space's low-variance directions - was tested and is FALSE. The hub reconstructs such a space essentially perfectly at every width examined. The cause is instead how similar the TARGET own items are to one another. Retrieval into a space means ranking that space vectors against each other, and in GPT-2 caption space every item sits at roughly +0.999 cosine from every other, so the distinctions that decide a ranking live in the last decimal place. The hub path adds one fitting stage through a bottleneck, and that error exceeds the gaps it must preserve. In a healthy target the items sit far enough apart that the same error is harmless - and the hub consensus effect can even improve on a directly-fitted map, which is why several pairs exceed 100 percent retention. A controlled reproduction on this project own vectors, collapsing a healthy target to the GPT-2 signature, moved retention from 152 percent to 31 percent with everything else held fixed.

What this adds to the claim. One 512-dimensional linear coordinate system serves seven encoders across four training objectives and two modalities, retaining 95.6 percent of pair-specific maps everywhere except when writing into a degenerate space, and doing so without measurable cost to the transfer result measured on five. For the deployed use - reading encoders out into a shared index - every encoder including GPT-2 is

usable, since GPT-2 reaches the bge readout space at 93.9 percent. The limitation is precise and it is a property of the target, not of the hub.

C.13.4 Two properties that are easily confused, and why bge-m3 remains the readout target

Reaching the hub and being written into are different properties. The hub reports, for each encoder, how well its linear map predicts hub coordinates - call it reach. It is tempting to read a low reach as predicting that the hub will reconstruct that space poorly. Across the seven registered spaces, it does not:

space	reach into hub (R2)	effective rank	writable into (R@1 retention)
BERT	0.211 - the LOWEST	65.0	76.6%
GPT-2	0.228 - higher than BERT	6.6	24.8%
SBERT	0.265	93.4	101.0%
bge-m3	0.288	97.4	98.1%
DINOv2-small / base / large	0.367 / 0.533 / 0.591	252.6 / 425.8 / 506.1	99.6% / 99.8% / 98.6%

Table C12e: BERT reaches the hub less well than GPT-2 and is three times more writable, so reach does not predict writability. Effective rank does.

The distinction matters because the two describe different halves of a round trip. Reach asks whether a space's information can get INTO the shared coordinates; writability asks whether the hub can produce vectors precise enough to RANK items once back in that space. The second is governed by how far apart the target's own items are: in GPT-2's caption space every item sits at roughly +0.999 cosine from every other, so a ranking turns on differences an extra fitting stage cannot preserve. A space can therefore be easy to reach and impossible to rank within, which is precisely the case that generated Table C12d.

Why bge-m3 remains the readout target. The caption head maps hub coordinates into a chosen text space, and on the measurements above bge-m3 is not the best available candidate. Sentence-BERT is marginally better on both relevant axes - 101.0 percent writable against bge's 98.1, and a higher raw R@1 in the ladder of Section C.14 (0.510 against 0.466). BERT, by contrast, is a poor choice on both: 76.6 percent writable and 0.362 raw.

candidate readout	writable into	raw R@1 (C.14)	verdict
bge-m3	98.1%	0.466	retained. Every ceiling, every percentage-of-ceiling figure and every cross-modal result in this report is measured against it; switching would make new numbers incomparable with Sections 4, C.8, C.10 and C.13 for a gain of about three points
SBERT	101.0%	0.510	marginally better, and available as an alternative in the demo notebook - but adopting it would cost the comparability above
BERT	76.6%	0.362	rejected on the measurements: weaker on both axes, and its masked-LM space is the middle rung of the ladder rather than a similarity space

Table C12f: the readout choice, stated with the numbers that argue against it as well as for it.

The honest summary is that bge-m3 is retained for comparability rather than because it won on merit. That is a defensible reason - a project whose every figure is quoted as a percentage of a bge-measured ceiling cannot swap the ceiling in its final section - but it should not be presented as though the data chose it. Section C.14's finding applies here too: what a good readout space supplies is a coordinate frame in which cosine is meaningful, and both contrastively trained candidates supply one, while the masked-LM and causal-LM spaces do not.

C.13.5 At what scale the encoders agree - and what that explains

Every similarity figure elsewhere in this report is a single summary of a pair. This section resolves the summary by SCALE, using nearest-neighbour overlap at k from 1 to 500 over 1,500 held-out items. Because chance overlap is k/N , the informative quantity is the ratio to chance, not the raw fraction: a raw overlap of 0.5 is strong at $k=10$ and barely above chance at $k=500$.

pair	k=1	k=10	k=100	k=500
bge - SBERT (highest)	967x	116x	11.4x	2.3x
img_base - img_large	777x	104.6x	10.1x	1.9x
img_base - SBERT (cross-modal)	236x	57.5x	7.6x	1.6x
img_base - GPT-2 (cross-modal)	70x	23.2x	3.9x	1.3x

Table C12h: neighbourhood agreement as a multiple of chance, for four representative pairs. Every one of the 21 pairs falls monotonically toward the chance floor as k grows.

The agreement is local. For all 21 pairs without exception the ratio to chance falls steeply with k , from hundreds-fold at $k=1$ to low single digits at $k=500$. The encoders concur strongly on what is NEAR what and diverge about global arrangement. This is the map-projection situation made quantitative: two charts of the same region preserve local adjacency while distorting global distance, and these encoders are charts of the same relational structure rather than copies of one global metric.

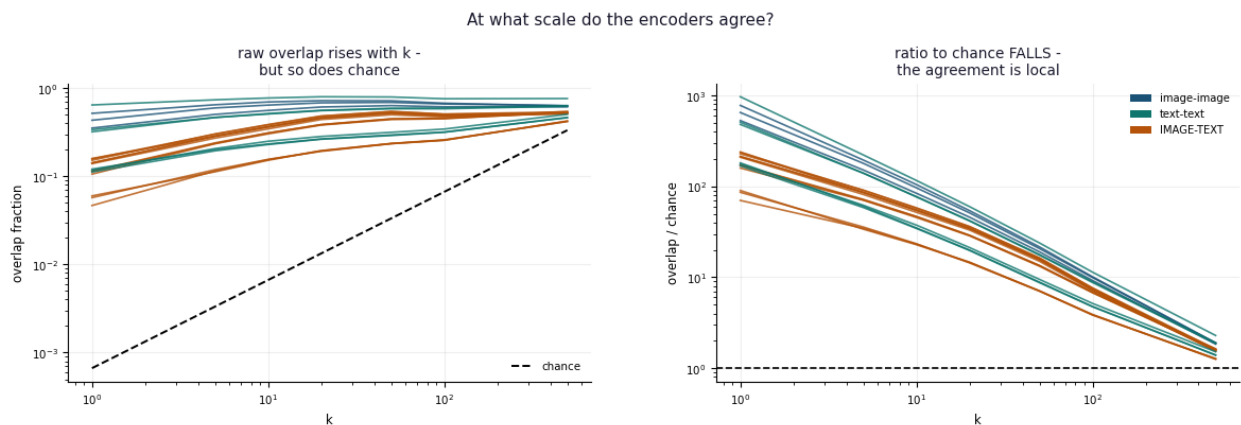


Figure C6a: raw overlap rises with k only because chance does (left); the ratio to chance falls for every pair (right), which is the signature of locally-shared, globally-divergent structure.

This resolves an apparent tension elsewhere in the report. Cross-modal rank correlation is modest (ρ about 0.29, C.11.2) while cross-modal retrieval through a fitted map is substantial (C.8). Both are correct because they read different scales: a linear map recovers what survives as GLOBAL linear structure, which is thin, whereas neighbourhood overlap reads the LOCAL structure, which is large. Retrieval sits between them because it samples both. The scale curve is why a linear adapter reaches most but not all of the ceiling: the part it misses is the local structure no single linear map can carry.

The local agreement is not uniform across items. Overlaying two encoders' neighbour graphs on one layout (Figure C6b) shows 68 of 238 edges surviving between DINOv2-large and SBERT - an image encoder and a text encoder sharing 29 percent of their neighbour graph with no map fitted. The per-item distribution is uneven: a mean of 2.3 agreed neighbours of 4 possible, only 3 percent of items sharing none, but a clear spread from items agreed everywhere to items barely agreed at all. Some items sit in the same relational position for every encoder; others are placed idiosyncratically. That subpopulation structure is a target for future work - the universally-agreed items are where cross-encoder transfer should be most reliable.



Figure C6b: dark edges are neighbour relations both encoders agree on; the layout is arbitrary (taken from one space) and only the edges are the measurement. Right: agreed-neighbour count per item is not spread evenly.

What explains the agreement (exploratory). Tagging each pair by modality, lineage and objective class and comparing group means at $k=10$ orders the three factors: objective (same 100.2x vs different 47.0x, gap +53.2) above lineage (95.0x vs 50.9x, +44.1) above modality (73.5x vs 44.9x, +28.6). The single highest-agreeing pair, bge and SBERT at 116x chance, shares a training OBJECTIVE and no lineage. Taken with the shape-agreement finding of C.13.3, this points the same way: what a model was trained to DO tracks representational agreement more closely than what it was built from. This is stated as exploratory description and nothing more - with 21 pairs the groups run as small as $n=1$, the pairs are not independent (three share a teacher), and no significance test is computed. Its value is in identifying which comparison merits a properly powered study: more encoders spanning OBJECTIVES, not more spanning sizes.

C.13.6 Local versus global structure, and which Platonic claim the evidence supports

The scale analysis of C.13.5 answers a question the rest of the project left implicit: is the shared structure the SAME at every scale, or only local? The answer is measured and unambiguous - it is local. Across all 21 encoder pairs the neighbourhood agreement, expressed as a multiple of chance, falls monotonically from hundreds of times chance at $k=1$ to low single digits by $k=500$. No pair holds flat. Local topology and global topology are therefore NOT the same: the encoders concur on which items are near which, and disagree increasingly about the wider arrangement.

scale	what it measures	typical ratio to chance	reading
local ($k=1$)	is my nearest neighbour your nearest neighbour	70-970x	strong shared structure
mid ($k=50$)	is my local cluster your local cluster	7-24x	agreement fading
global ($k=500$)	is my third of the space your third	1.3-2.3x	near chance - little shared

Table C12i: agreement is a function of scale, not a constant. The local and global regimes differ by roughly two orders of magnitude.

Which Platonic claim this supports - and which it refutes. The Platonic Representation Hypothesis is often stated loosely, as though independently trained models converge to the SAME embedding space. In that strong form this project's evidence refutes it: raw cross-space cosine is zero, and global neighbourhood agreement is near chance. Two models do not build the same space.

The precise form the evidence SUPPORTS is different and weaker: independently trained models are different CHARTS of one shared relational manifold. Each encoder f_i maps the underlying structure of the world into its own coordinate system; the coordinate systems are unrelated - hence zero raw cosine and the need for a per-encoder map - while the local relations survive the change of chart. The map-projection analogy is exact rather than decorative: two projections of the Earth preserve local adjacency and distort global distance and area, and these encoders do the same. What is shared is the territory; what differs is the projection.

This reading is consistent with every result in the project and resolves what would otherwise look like a contradiction. The linear maps of Sections 4 and C.8 work because SOME structure survives as global linear form - enough to retrieve at most of the ceiling. They fall short of the ceiling because the richest agreement is local, and no single linear map transports local neighbourhood structure globally. The unsupervised recovery of C.12 succeeds because Gromov-Wasserstein reads exactly the local distance structure that is most strongly shared. And the capacity result of C.10 - alignment rising with scale - is the Platonic prediction in its testable form: better models are better charts of the same manifold, so their local agreement grows. The naive claim is too strong to be true; the chart claim is precisely as strong as the measurements, and no stronger.

Bounds, stated with the claim. The manifold is inferred from seven encoders on one image domain and one language; a shared chart of COCO in English is not proof of a universal one. The causal reading - that the shared structure is world-induced rather than induced by common architecture or objective - is supported by cross-modal and cross-lineage agreement surviving (C.13.2, C.13.5) but is not established, since the strongest agreement still tracks shared training OBJECTIVE (C.13.5) and the encoders are not architecturally independent until a convolutional one is added. The evidence narrows the hypothesis to its defensible core; it does not settle its cause.

C.13.7 Characterising the spaces: intrinsic dimension, near-isometry, and shared subspace

Three closed-form measurements (Experiment G6) describe WHAT KIND of objects the encoder spaces are and HOW two of them relate. None strengthens the global-similarity claim - Section C.13.5 measured global agreement as weak, and these characterisations are consistent with that rather than a route around it. Their value is precision: they turn two words the project had been using informally - manifold and 'related, not rigid' - into measurements, and they corroborate the local-structure story by three independent routes.

Intrinsic dimension. The TwoNN estimator reads the effective degrees of freedom of each space from nearest-neighbour distance ratios. All seven spaces sit between 11.2 and 19.9 dimensions inside ambient widths of 768 to 2048 - under 3 percent of the storage in every case. This is the measured basis for the word manifold: the semantic structure occupies low tens of dimensions inside a space 40 to 180 times wider. It is a per-space property and says nothing by itself about whether two spaces share that structure.

space	ambient	intrinsic d	% of ambient
img_small	768	19.9	2.6%
img_base	1536	17.5	1.1%
img_large	2048	16.1	0.8%
txt_bge	1024	12.9	1.3%
txt_gpt2	768	13.5	1.8%
txt_bert	768	16.3	2.1%
txt_sberr	768	11.2	1.5%

Table C12j: intrinsic dimension (TwoNN). Thin everywhere, regardless of ambient width.

Near-isometry across all 21 pairs. Orthogonal Procrustes - the best pure rotation between two spaces - was previously measured for one pair only (Experiment B, 0.038). Run across all 21 pairs, it stays low while ridge stays high (mean 0.194 vs 0.667): the spaces are related by more than a rotation, differing in stretch, across the whole set. This confirms Finding 2 broadly rather than from one case. Two details matter. First, cross-modal and GPT-2 pairs give NEGATIVE Procrustes - a rotation cannot even begin to align them - which is the sharpest form of 'related, not rigid'. Second, one pair is the exception: bge-m3 and SBERT reach 0.758, near-isometric - and it is the same same-objective text pair that tops the shape-agreement table (+0.764) and the CCA count below. Every method flags the same pair as most alike.

These Procrustes values are measured after reducing both spaces to a common PCA dimension, which equalises per-axis variance and performs part of the alignment. They are therefore systematically higher than, and NOT comparable to, Experiment B's raw 0.038 (raw 512-d into raw 768-d, no PCA). The raw-space near-isometry figure remains 0.038; these characterise the pairs on a common footing.

Shared versus private subspace (CCA). Canonical correlations, whitened and scored on held-out rows, count how many directions two spaces genuinely share. The count orders exactly as every other measure in the project:

relationship	example pair	dims corr>0.5 (of 64)	reading
image siblings	img_base - img_large	64	nearly all shared
same-objective text	bge - SBERT	64 (54 above 0.9)	the most-shared pair in the project
mixed-objective text	GPT-2 - SBERT	64 (few above 0.9)	shared subspace, thinner core
CROSS-MODAL	img_base - txt_bge	6	a thin shared subspace

Table C12k: CCA shared-direction counts. Cross-modal pairs share the fewest directions (6-32 of 64); image siblings and same-objective text pairs share nearly all.

Why this corroborates rather than extends. Three methods that share no machinery - a neighbour-distance dimension estimate, a rotation fit, and a whitened correlation analysis - place the pairs in the same order that kNN overlap (C.13.5) and rank correlation (C.11.2) already did: image siblings and same-objective text pairs

most alike, cross-modal pairs least. That a fourth and fifth independent route reproduce the ordering is the strongest kind of internal check. And the one pair every method singles out as near-isometric - bge and SBERT, sharing an objective and no lineage - is the same pair Section C.13.5 flagged, which is why the causal reading there (objective tracks agreement more closely than lineage or modality) survives this additional scrutiny. None of it raises the global-similarity number, and it was not meant to: the structure is real and local, thin and global, and now characterised.

C.13.8 Calibration against a permutation null, and replication on the literature's metric

Gröger, Wen and Brbić (ICML 2026) argue that representational-similarity metrics are confounded by network scale: wider embeddings produce a positive similarity baseline even between independent representations, so raw scores can rise with model size without any convergence. They propose replacing the analytic baseline with a permutation null and report that, once calibrated, global spectral convergence largely disappears while local neighbourhood agreement survives. Their conclusion - that representations converge to shared local neighbourhood relationships - is the same one Section C.13.5 reached independently. This section applies their correction to this project's own measurements.

The test. One side's rows are shuffled, destroying the item correspondence while leaving both spaces' widths, spectra and internal geometry untouched. Whatever the metric still reports is the scale-and-geometry baseline. Twenty shuffles per pair, three metrics, all 21 pairs.

metric	mean permutation null	effect on this project
k-NN overlap at k=10	+0.016, which is exactly the analytic chance rate k/N	none - the ratio-to-chance used in C.13.5 was already calibrated. Set intersection carries no width bias
distance rank correlation	+0.000	none - the null is zero as assumed
CKA	+0.094 , and it rises with embedding width (Spearman +0.56)	real - raw CKA overstates agreement, most for the widest pairs

Table C12l: the confound is metric-specific. The project's primary local measure is unaffected; CKA requires calibration.

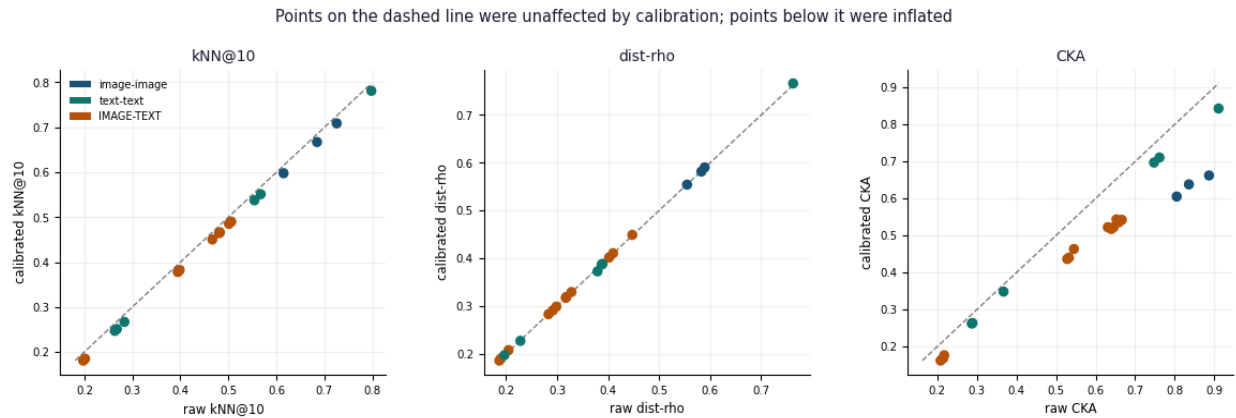


Figure C6c: distance below the diagonal is the scale confound. k-NN and rank correlation sit on the line; CKA sits below it, furthest for the widest pairs.

The correction strengthens the project's causal reading rather than threatening it. Because the null grows with width, the two DINOv2 siblings - the widest pair in the set at 1536 and 2048 dimensions - were inflated most. Calibration removes 0.225 from their raw CKA of 0.887, while the bge-SBERT pair, narrower and sharing a training objective rather than a lineage, loses only 0.068 from 0.911. The gap between the objective pair and the lineage pair widens from +0.024 raw to **+0.181 calibrated**, a factor of seven and a half. The scale confound had been working AGAINST the finding of Section C.13.5 that shared objective tracks agreement more closely than shared ancestry; correcting it makes that finding stronger.

Where this project's data differs from theirs. They report that agreement about local distances does not survive calibration while neighbourhood agreement does. Here both survive: every pair clears its null on rank correlation as well as on neighbourhood overlap. The ORDERING agrees - neighbourhood z-scores run 101 to 429 while distance z-scores run 12 to 127, so neighbourhoods are consistently the more robust invariant - but the distance measure is not extinguished at this scale. Reported as a partial replication: the direction reproduces, the strength does not.

Replication on the literature's own metric. CKNNA (Huh et al. 2024) is CKA restricted to mutual k-nearest neighbours, recovering plain CKA as k approaches n. Computed across k for all 21 pairs, **21 of 21 score**

higher locally than at the global limit, and the fall is steepest for cross-modal pairs - image-to-GPT-2 drops from 0.917 at $k=10$ to 0.169 at the limit. Ranking the pairs by CKNNA at $k=10$ and by neighbourhood overlap gives a rank correlation of **+0.81**. The local-versus-global conclusion therefore does not depend on which instrument was chosen: three metrics, two of them from the literature this project is testing, order the pairs the same way and reach the same verdict.

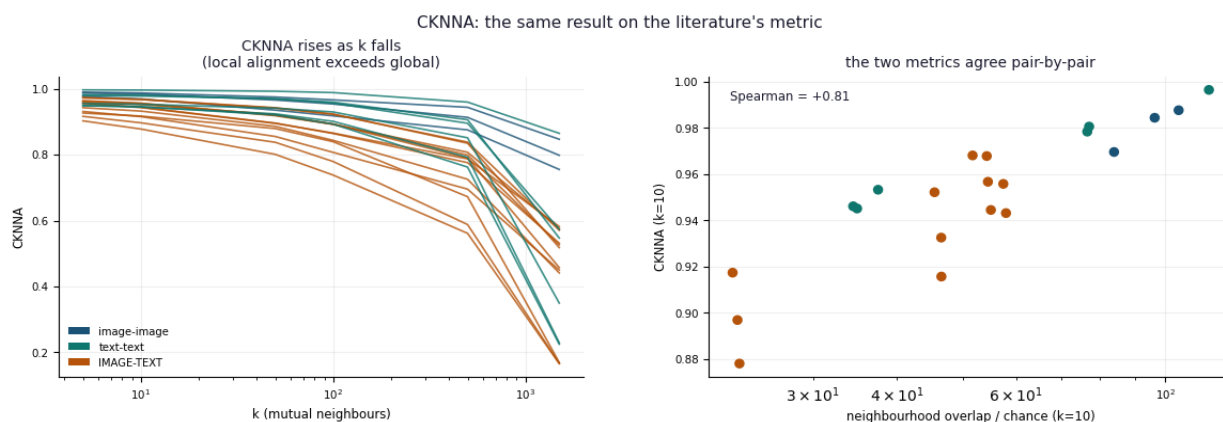


Figure C6d: CKNNA rises as k falls for every pair (left); the two metrics agree pair-by-pair at Spearman +0.81 (right).

C.13.9 A correction: what the ratio-to-chance curve does and does not show

Section C.13.5 presented the fall of neighbourhood overlap relative to chance - from hundreds of times chance at $k=1$ to roughly twice chance at $k=500$ - as the evidence that agreement is local. That presentation was **partly wrong, and this section corrects it** rather than removing it, in keeping with the project's treatment of its other falsified explanations.

The problem. Ratio to chance is bounded above by N/k . At $k=1$ with $N=1500$ the maximum achievable ratio is 1500; at $k=500$ it is only 3. A pair with PERFECT neighbourhood agreement at every scale would therefore still show a falling ratio. Part of the observed decline is the shrinking ceiling, not shrinking agreement.

The correct diagnostic. A chance-corrected agreement statistic of Cohen's kappa form - $(\text{observed} - \text{chance}) / (1 - \text{chance})$ - is bounded in $[0,1]$ at every k and is therefore comparable across scales. Recomputed that way, the picture is different:

pair	k=1	k=10	k=50	k=100	k=500	shape
bge - SBERT	0.645	0.771	0.787	0.741	0.644	rises, peaks $\sim k=50$, falls back
img_base - img_large	0.518	0.695	0.709	0.652	0.445	same shape, steeper fall
img_base - txt_bge (cross-modal)	0.141	0.360	0.507	0.449	0.268	same shape, largest rise
img_base - txt_gpt2 (cross-modal)	0.046	0.149	0.209	0.204	0.131	same shape, lowest throughout

Table C12m: chance-corrected neighbourhood agreement. The curve is not monotonically falling - it rises to a peak around $k=20$ to 50 and declines thereafter.

What this changes. The k -NN ratio-to-chance curve on its own does NOT establish that agreement is local. Under a bounded normalisation the strongest agreement sits at intermediate neighbourhood sizes, not at $k=1$, and the decline at large k is real but smaller than the ratio suggested. The claim in C.13.5 was overstated in its evidence, though not in its direction.

What still stands, and why. The local-over-global conclusion rests on the CKNN sweep of C.13.8, which is not subject to this artifact: CKNN is bounded in $[0,1]$ at every k , is the same metric family throughout the sweep, and recovers plain CKA at the limit. On that measure **21 of 21 pairs score higher locally than globally**, with cross-modal pairs falling furthest (image to GPT-2: 0.917 at $k=10$ against 0.169 at the limit). Two further independent lines agree: raw cross-space cosine is zero while neighbourhood overlap is far above chance, and Gröger et al. (2026) report the same local-survives, global-weakens pattern after permutation calibration on 204 model pairs. The conclusion is unchanged; one of the four arguments for it was faulty and has been replaced by the three that are not.

Why the peak at intermediate k is itself interesting. Chance-corrected agreement is highest at k of roughly 20 to 50 for every pair measured. At $k=1$ the measure is dominated by which single item happens to be closest, which is noisy; by $k=500$ it is asking about a third of the corpus, where the spaces genuinely diverge. The peak marks the scale at which the shared structure is best expressed - neighbourhood-sized, not point-sized and not global. That is a sharper statement than the original claim and it emerged from correcting it.

C.13.10 What the hub does to local structure — and why it works at all

Two halves of this project had not been made to meet. The science says what encoders share is LOCAL: neighbourhood-scale agreement is large and global agreement is thin (C.13.5, C.13.8). The engineering is a GLOBAL linear projection: whitened PCA plus one ridge map per encoder. A linear map recovers global linear structure, which is the thin part — so why does the hub deliver 93 to 96 percent component transfer at all? Experiment G7 measures it.

The map is not gentle. Comparing each encoder's native neighbour sets with its neighbours after mapping into the hub gives a mean overlap of 0.594 at $k=10$ — the projection substantially reshapes every encoder's local structure rather than rotating it. GPT-2 is the extreme case at 0.257, the lowest of the seven.

The decisive measurement, stratified. Comparing cross-encoder neighbourhood overlap before and after mapping gives a mean gain of +0.111 across all 21 pairs — but that average is not interpretable, because it mixes two different phenomena. Splitting by the project's own collapse diagnostic (mean pair-cosine, which unlike effective rank does not depend on ambient width) separates them cleanly:

population	mean change in kNN overlap	n	reading
healthy spaces only (pair-cos < 0.30)	+0.010	6 (2 negative)	the hub is NEUTRAL on local agreement
pairs including a degenerate space	+0.151	15	large gains, concentrated on the worst-conditioned spaces
all pairs (not interpretable)	+0.111	21	the average of two different effects

Table C12n: the aggregate hides the structure. Spearman between a pair's worst pair-cosine and its gain is **+0.82** — the improvement is driven by how degenerate the pair's weaker space is, not by hub alignment.

The answer to the question. On healthy spaces the hub neither creates nor destroys local agreement; it relocates it into shared coordinates. Transfer therefore runs on the GLOBAL linear component — real, if thin — while the rich local structure measured in C.13.5 is carried along intact but unimproved. This resolves both halves of the puzzle at once: the hub works because the global component genuinely exists, and it never reaches the ceiling because the local structure that carries most of the shared information is not what a linear map transports. What had been two separate observations — a thin global signal and a persistent retrieval shortfall — are the same fact seen twice.

The C.11 rescue, for the fourth independent time. The large gains belong entirely to pairs involving a degenerate space, and GPT-2 is the clearest case: the hub destroys its native neighbourhoods more thoroughly than any other encoder's (preservation 0.257) and its cross-encoder agreement improves most (up to +0.417). That is not a contradiction. Its native neighbourhoods were close to arbitrary — every item at +0.999 cosine from every other — so reshaping them is an improvement rather than a loss. The hub's whitening repairs a degenerate coordinate frame exactly as the explicit whitening of C.11 did, and as the hub-basis rescue of C.13.3 did. It is a property of the target's geometry, not of the hub aligning local structure, and it must not be averaged into the headline.

Bounds. The healthy population is six pairs, two of them negative, so 'neutral' means 'no detectable effect at this sample size' rather than a demonstrated null. A first version of this analysis used effective rank divided by ambient width as the collapse flag and wrongly classified SBERT — whose pair-cosine of +0.136 makes it the healthiest text space in the project — as degenerate, because text encoders legitimately carry lower rank-to-width ratios than image encoders. That version reported the opposite verdict. The flag was changed to pair-cosine, which is width-independent, and the misclassification is recorded here rather than silently fixed.

C.13.11 The width cliff reproduced, and two explanations falsified

Section C.13 reported that hub width forces a choice: narrow and components transfer, wide and transfer collapses while reconstruction improves. Experiment G7 sweeps hub width, separates two readouts that are easy to conflate, and tests both candidate explanations of the trade-off. The cliff reproduces. Neither explanation survives.

HUB_DIM	self / reconstruction	zero-shot / portability	step	amplification $1/\lambda$
64	0.325	0.350	—	0.25
128	0.382	0.435	+0.085	0.35
256	0.412	0.474	+0.039	0.51
512	0.420	0.489	+0.015	0.74
768	0.411	0.312	-0.177	0.93

Table C12o: only portability cliffs. Reconstruction plateaus, because feeding an encoder's own coordinates through a wider hub can only retain more information.

A measurement error, corrected. The first version of this sweep fitted the head on one encoder and evaluated it on the SAME encoder. That measures reconstruction, not portability, and cannot cliff by construction — which is exactly what it showed, a plateau at 0.412, 0.420, 0.411. The cliff appeared only once the readout matched the one Section C.13 had used. The failure to reproduce was a fault in the instrument, not evidence against the phenomenon.

First explanation: local-structure loss. FALSIFIED. The natural hypothesis was that wider hubs preserve less neighbourhood structure and therefore transfer less well. Across the cliff step, neighbourhood preservation moves by -0.018 and cross-encoder overlap by -0.030 — the same magnitude as their moves elsewhere in the sweep — while transfer drops 0.177, six times more. Cross-encoder overlap falls smoothly and monotonically across the whole sweep while transfer rises and then collapses. **A smooth predictor cannot explain a discontinuous outcome.**

Second explanation: spectral amplification. ALSO FALSIFIED, by the same argument. Whitened PCA divides each direction by its singular value, so the natural account was that past some width the hub admits directions whose variance is small enough that whitening turns them into noise carrying unit weight. That account predicts a threshold, which is the right shape — but the amplification factor does not have one. It rises 0.25, 0.35, 0.51, 0.74, 0.93 across the sweep, and its growth rate **slows** at the cliff ($1.45\times$ per step before, $1.26\times$ across it). Transfer collapses precisely where amplification is decelerating. Cumulative variance is smooth too. The same objection applies, and this project cannot make an exception for its own preferred explanation.

What this left, at the time of writing. The cliff was real, reproduced, and unexplained: both candidates predicted smooth degradation where the phenomenon is a discontinuity. It did not stay that way - the resolution is recorded below, and in full in E.8, in the order it happened. **The falsification's boundary, since widened.** Only one operationalisation was tested here: the amplification applied to the last direction each width admits. G8 later tested four more - accumulated noise energy, noise-to-signal ratio, condition number, shared-direction fraction - on a pre-registered discontinuity standard; all four fail the same way, smooth where the outcome is a step (E.3).

One boundary on the second falsification, since closed. Seven accounts were falsified and the eighth is CONFIRMED (Appendix E.8): the cliff is least squares dividing by a near-null singular value, which appears exactly when hub width reaches the source encoder's rank. It is a numerical artifact of the fit, not a property of shared coordinates.

And a second confirmation of C.13.10. Across the sweep, cross-encoder local agreement falls monotonically from 0.603 to 0.462 while zero-shot transfer rises from 0.350 to 0.489 before the cliff. The two move in opposite directions over the whole usable range, so transfer cannot be running on local structure: it improves precisely where local agreement degrades. That is independent evidence, from a different measurement, for the conclusion of C.13.10.

C.14 The encoder ladder: Section C.11 as a pre-registered prediction

Section C.11 explains the GPT-2 arm's collapse and rescue with one mechanism: what similarity training supplies is isotropy, not correspondence. An explanation fitted to the data that motivated it is weak evidence; the same explanation predicting the outcome for encoders it has never seen is strong evidence. Two such encoders were therefore run, with the predictions registered in the notebook before the numbers existed: BERT (masked-language modelling, bidirectional - predicted to land BETWEEN GPT-2 and the contrastive encoders in both effective rank and raw retrieval) and Sentence-BERT all-mpnet-base-v2 (contrastive - predicted near the bge arm, with little to gain from whitening). Image side, protocol, caption averaging, and the whitening recipe are identical to Sections C.9 and C.11: the cached DINOv2-base cls+patch embeddings, 18,066 pairs at 11.1 rows per input dimension, transforms fitted on training targets only.

encoder	training objective	eff. rank (of 768)	mean pair-cos	raw R@1	whitened R@1
GPT-2 (C.9/C.11)	causal LM	6.5	+0.999	0.160	0.478
BERT	masked LM, bidirectional	65.0	+0.810	0.362	0.521
SBERT	MLM + contrastive	93.4	+0.139	0.510	0.588
bge-m3 (reference)	heavy contrastive	high	low	0.466	-

Table C14: the ladder. All rows are targets for the same image encoder under the same protocol; only the text encoder's training objective varies.

Prediction 1 - confirmed exactly. Raw retrieval orders by the target space's effective rank with rank correlation **+1.00** across the ladder: 6.5, 65.0, 93.4 in effective rank maps onto 0.160, 0.362, 0.510 in R@1. BERT landed precisely where registered - between GPT-2 and the contrastive encoders - and the severity ordering of the anisotropy itself came out as predicted (pair-cosine +0.999, +0.810, +0.139: degenerate, anisotropic, near-isotropic). Raw cross-modal retrieval against a language-model target is governed by the target's geometry, not by the amount of language understanding its objective implies.

Prediction 2 - partial, and the miss is the sharper finding. Whitening was predicted to collapse the ladder onto one level. It lifted every arm - 0.478, 0.521, 0.588 - but a spread of 0.110 remains, and the residual ordering is IDENTICAL to the raw ordering. The refined statement replaces C.11's original one: **isotropy is necessary but not sufficient**. Whitening equalises how many directions carry variance - all three targets reach effective rank near 765 afterwards - but it cannot install WHICH directions carry meaning. Contrastive training contributes two separable things: a coordinate frame (isotropy), which a label-free linear transform supplies for free and which accounts for most of the gap, and a semantic arrangement of directions, which it cannot. That residue is why a purpose-built contrastive encoder still leads after every arm is whitened.

What this does to the project's claims. C.11's mechanism survives its first out-of-sample test and gains a boundary. The quantitative split is worth stating: for GPT-2, whitening closes the distance from 34 percent of the bge arm to 103 percent - the frame was nearly the whole story for the most degenerate space. For BERT and SBERT the lift is smaller and a training-objective residue remains visible. Reported failures of cross-model alignment should still be checked against an isotropy battery first - but after the battery, what remains is a genuine signature of the training objective, and the ladder measures its size.

Appendix D — A derived capability: cross-tier membership and similarity

This appendix extends the engineering discussion of Section 4.5 with a capability the shared index provides at no additional cost, identified after the main write-up.

Because the adapter places phone-side vectors in SigLIP's coordinates, the shared index supports **image-as-query** operations across tiers, not only text search. Computing $\hat{u} = L2(v \cdot W)$ for a photo on the device and querying the index with $\hat{u}$ itself yields, from the top of the ranking: **(a) membership and near-duplicate detection** — if the same photo, or a crop or re-save of it, is indexed on the server, its native SigLIP vector returns with a cosine far above the background, and thresholding on the gap (the triage rule of the B5 sandbox) answers 'is this already in the library?' without uploading the photo; and **(b) cross-tier similar-image search** — lower cosines with a clear ranking give 'server photos like this one', and symmetrically for server photos against phone-indexed content.

Two boundaries keep the claim precise. First, this predicts an image's *address* in SigLIP's space, never its content: W maps 512 numbers to 768 and cannot reconstruct pixels (though, per the security discussion in Appendix B, embeddings should still be treated as content-bearing, since dedicated inversion models recover approximate content from them). Second, the capability exists only through the adapter — a raw MobileCLIP vector compared against the server gallery scores exact chance (Section 4.3), so cross-tier membership is a property the matrix creates, not one the encoders share natively.

The expected accuracy is bounded below by the measured text-query results (Section 4.3), and image-to-image retrieval within one contrastive space is typically easier than text-to-image, since no modality gap is crossed. The measurement is a three-line variant of stage B3 — querying the SigLIP gallery with the adapted vectors instead of the caption embeddings — and is included in the B3 notebook as an optional cell.

D.2 Three decision problems on the shared index, and where a threshold belongs

Appendix D reports the membership capability with a per-query statistic: the true twin outranks the best impostor for 98.7 percent of queries. That number is correct for what it measures, but 'membership' can be read as a deployable detector - a system that answers yes or no from a score against a cutoff - and that is a different, harder question. This section separates three decision problems that the same adapter and the same 1,000 held-out vectors support, and measures each with the instrument appropriate to it.

Problem	Form	Question	Instrument	Result
Verification	1:1	are these two vectors the same photo?	ROC / AUC	0.999
Ranking	1:N, per query	does the twin outrank every impostor for THIS query?	positive-gap rate	0.987
Membership	1:N, one global cutoff	does a single threshold separate indexed from absent?	ROC / AUC	0.889

Table D2: the same 0.79 MB adapter measured as three different deciders. Raw MobileCLIP without the adapter scores AUC 0.510 on membership and 0.477 on verification - chance - so the entire capability in every row is created by W.

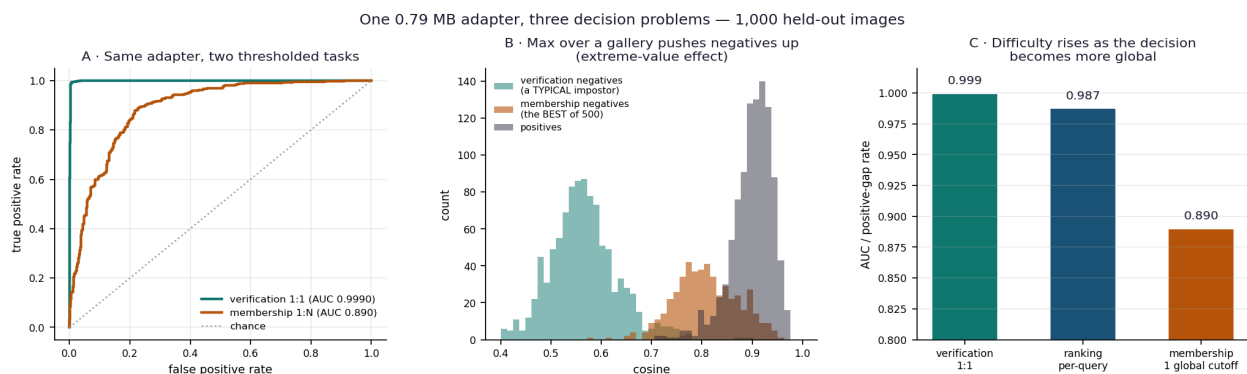


Figure D2: the two ROC curves, the score distributions that explain their difference, and the difficulty ordering.

Why difficulty rises left to right. A verification negative is a typical impostor, and typical impostors score around 0.57 - far below the positives at 0.90, so a threshold separates them almost perfectly. Membership must instead reject the BEST impostor among 500: the maximum of 500 draws sits far into the tail, averaging 0.809, and the distributions overlap. Ranking sits between the two because it also faces the best impostor but only per query - each query needs its twin to win locally, not to clear a cutoff shared with every other query. Cosine scales vary across queries, so one global threshold cannot be optimal for all of them.

Operating points for the deployed system. At a false-positive rate of 1 percent the membership detector recovers 14.2 percent of true members; at 5 percent, 45.0; at 10 percent, 61.4. The best F1 is 0.840 at a cosine cutoff of 0.856. The honest deployment statement is therefore: cross-tier ranking is near-ceiling, verification of a claimed match is near-perfect, and duplicate DETECTION with one global threshold is good but materially harder - a consumer should prefer per-query comparisons wherever the application allows.

A note on scope. ROC applies to these tasks because each has a binary label and a score. It does not apply to R@K retrieval, which has one correct answer in a gallery and no threshold - the cumulative rank curve is the analogous instrument there. One further observation ties this appendix to Section C.11: in caption verification through the adapter chain, positive cosines average just 0.136 against negatives at -0.024, yet AUC is 0.997 - absolute similarity values are small while the decision is nearly perfect, because AUC measures separation, not magnitude. Which metric is informative depends, again, on the question and the geometry, not on the size of the numbers.

D.3 Hubness: a second pathology the adapter removes

Anisotropy - variance concentrated in few directions - is the geometric property this project measures throughout. Hubness is a distinct one: a few gallery items become the nearest neighbour of disproportionately many queries. It is a property of the neighbourhood graph rather than of the variance spectrum, it grows with dimension, and it matters here because a hubbed space can pass verification while failing retrieval - a handful of hub vectors outranks the true partner for many queries at once. The standard statistic is the skewness of N_k , the number of times each gallery item appears in some query top-k list.

Measured on the same 1,000 held-out images with $k = 10$, where perfectly even usage would give every item a count of 10:

query space	N10 skewness	top 1% of items absorb	most-hubbed item	never a neighbour
adapted (phone through W)	1.61	4.6%	71 queries	0.1%
raw (no adapter)	4.28	10.9%	202 queries	18.2%
ceiling (SigLIP native)	1.41	3.9%	65 queries	0.0%
random-direction queries	0.91	2.9%	38 queries	0.1%

Table D3: hubness of the cross-tier index. The random-query row is a floor, not a target - unstructured queries spread evenly because they carry no signal.

The untranslated cross-tier space is severely hubbed: one image is the nearest neighbour of 202 of 1,000 queries while 18.2 percent of the gallery is never anyone neighbour at all. The adapter removes almost all of it, landing at 1.61 against the jointly-trained ceiling 1.41. This adds something the retrieval and AUC figures do not show on their own: the 0.79 MB map does not merely rotate one space onto the other, it restores a healthy neighbourhood structure in which the index is used evenly. For a deployed shared index that property matters directly - a hubbed index wastes most of its entries and returns the same few items to unrelated queries.

Read together with Appendix D.2, this completes the diagnostic set for a retrieval failure. Effective rank and mean pair-cosine describe how the variance is spread; N_k skewness describes how the neighbourhoods are shared; and the gap between verification AUC and $R@1$ says whether what remains is coarse correspondence without pair-specific resolution. A collapse in $R@1$ alone does not establish that shared structure is absent - Section C.11 is the decisive counterexample, where retrieval had collapsed to 0.130 while held-out R -squared stood at 0.414, and a label-free linear transform recovered it to 0.478.

D.5 Artifact certification: 47 automated checks

Every stored artifact this project draws on was put through an automated battery in a single run, after all experiments were complete. The battery is notebook **H1** (power diagnostics); the per-space geometry measures it reports are also available standalone, for any $[n, d]$ representation, in notebook **V1** (vector diagnostics). Both are tooling rather than experiments - they certify measurements rather than producing findings - which is why they carry letters outside the A-G experiment series.

The checks are ordered by triage value: statistical power first, then row alignment, then artifact sanity, then target geometry, then cache-key integrity. The battery distinguishes two outcomes deliberately. A **FAILURE** means the artifact itself is suspect and nothing downstream of it should be believed. A **FLAG** means the check found a real property of the data rather than a defect - a geometry that changes how results must be read, not one that invalidates them.

check	what it catches	result across all artifacts
rows per input dimension	starved fits, which bias R-squared downward and can fake a negative trend	PASS - every fit between 5.9 and 9.8 rows/dim, above the pre-registered floor of 5
alignment (shuffle test)	silent row misalignment, the classic pipeline bug	PASS - honest fits beat shuffled ones by 0.545 to 0.863, against a 0.2 threshold
artifact sanity	NaN/Inf, zero rows, duplicates, broken normalisation	PASS - normalisation exact to $1e-7$; 20 duplicate rows in the caption array, consistent with repeated COCO captions
target geometry	a collapsed or hubbed target - the most likely cause of a WRONG negative result	2 FLAGS , both the GPT-2 caption space (eff. rank 6.5/768, pair-cos +0.999); all other targets pass
cache-key integrity	a checkpoint whose contents contradict the tag in its filename	PASS - all three DINOv2 caches match their declared encoder and pooling

Table D4: the battery, run across Experiment A activations, Experiment B pairs, all three DINOv2 caches and every cross-modal pairs file. 47 checks, no hard failures.

Two results deserve comment. First, the cache-key check passing on all three DINOv2 caches retires a specific risk: several corrections during this project traced to a cache whose filename promised one configuration and whose contents were another, and the check now verifies content width against the encoder named in the filename. Second, the two geometry flags are not merely expected but informative: the battery, run blind across the whole directory, independently reproduced the Section C.11 signature - effective rank 6.5 of 768 and mean pair-cosine +0.999 - in two separately generated GPT-2 files. That is a reproducibility check the project did not previously have.

V1 was run separately on the Experiment B pair and reproduces, from an independent code path, three figures reported elsewhere in this document: verification AUC 0.9994 against the 0.999 of Section D.2, per-query ranking $R@1$ 0.987 against the same, and a 69-fold spread in the adapter's singular values against Section 4. It also localises the anisotropy: the vision spaces carry a large shared **MEAN DIRECTION** (mean-direction ratio 0.427 and 0.731) over an otherwise reasonably spread body - effective ranks 140.9 of 512 and 181.7 of 768 - which is why centering alone removes their pair-cosine, and why they never needed the whitening that Section C.11's collapsed target did.

One honest qualification. The battery certifies the **MEASUREMENT**, not the science. It cannot tell whether a pooling or preprocessing choice was the right one, only whether the resulting arrays are well-formed, aligned, powered and geometrically legible. And row alignment **ACROSS** different encoders cannot be proved from the vectors alone - it follows from the caches having been built over the same deterministic identifier list, which the extraction notebooks enforce and this battery assumes.

D.4 The allegory, measured

The hypothesis this project tests carries a two-thousand-year-old picture with it. In Plato cave, prisoners chained before a wall take the shadows cast on it for reality; only a prisoner who is turned sees that the shadows were of solid forms all along. The Platonic Representation Hypothesis borrows the name deliberately: independently trained models are the prisoners, their raw vector spaces the shadows, and the shared structure of the world the forms that both are shadows of. This project adapter is the turning - not new sight, and not a new model, but a single linear map that rotates one encoder shadow until it falls on the other, revealing that the two were casts of one reality.

The figure makes the allegory quantitative. Read as a decision - can a threshold tell a true correspondence from a false one - the three prisoners separate cleanly. Chained (raw MobileCLIP, no adapter) the shadows carry no recoverable signal and the curve sits on the diagonal, AUC 0.481. Turned (the 0.79 MB adapter) verification reaches AUC 0.999: the two encoders name the same forms almost perfectly. Membership under a single global cutoff - the harder act of reading every shadow by one standard - reaches 0.889, the middle curve. The turning is cheap, linear, and touches no pretrained weight; what it reveals is that the agreement was always there to be found.

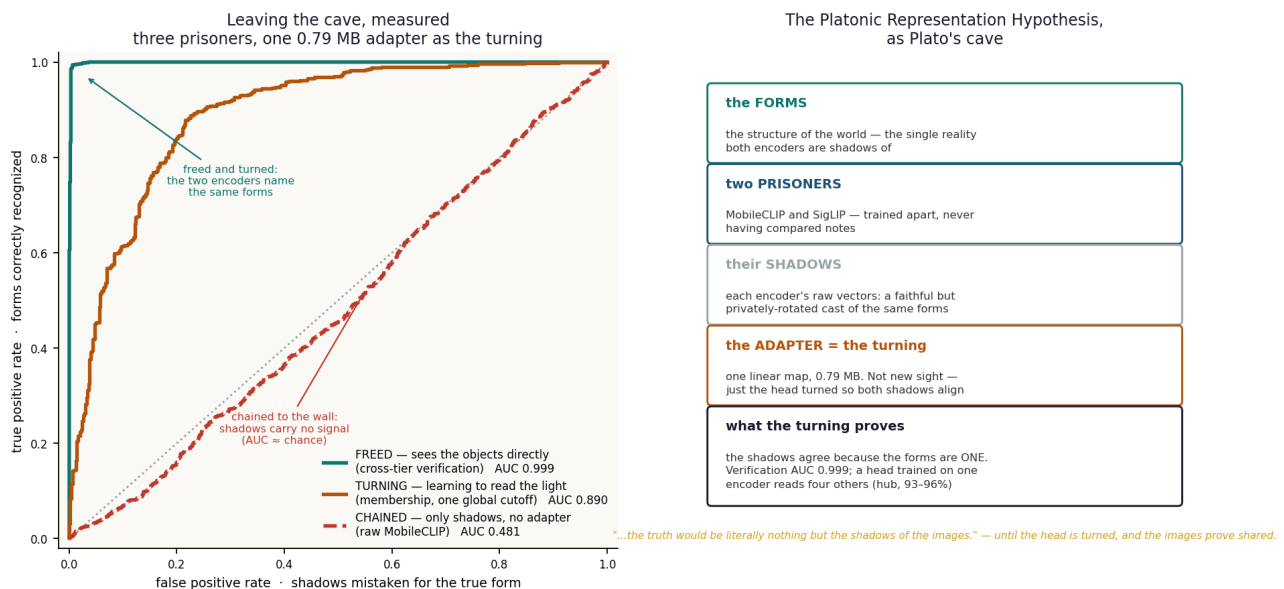


Figure D4: the cave allegory as an ROC. Left, three prisoners as three decision problems on real held-out data. Right, the correspondence between the allegory and the measured system.

Appendix E - K-series addenda (23 August 2026)

Four measurements made after the body of this report was written. They are recorded as an addendum rather than folded into the text so that what was measured originally and what was measured later stay distinguishable.

All four ran against a hub RECONSTRUCTED from the surviving cached encoder spaces, because the original G-series artefacts were not persisted. Section E.1 states what that reconstruction does and does not reproduce, and every later section is bounded by it.

E.1 The reconstructed hub, and what it is good for

The hub basis, per-encoder maps and singular spectrum from Appendix C.13 were not written to durable storage and did not survive. They were rebuilt from the cached encoder spaces, which did.

The rebuild was gated on the published singular values rather than accepted on faith. A raw (non-normalised) concatenation reproduces the published spectrum SHAPE to 0.71 percent across widths 128 to 768, after fitting one global scale of 0.846. That scale is not a discrepancy to explain away: whitened PCA divides by the square root of the eigenvalue, so scaling the input by c scales the eigenvalues by c^2 and the whitening by $1/c$, and the two cancel exactly. Hub coordinates are invariant under a global rescaling of the input.

Three details are worth recording because each was initially wrong. The hub is built on RAW features - an early attempt normalised rows first and missed the published spectrum by 97 percent, the ratio being the mean row norm rather than the square root of the sample size. The roster is resolved by explicit (file, key) pairs, not by substring: substring matching silently pulled the image side out of two cross-modal files and duplicated two encoders. And the scale must be fitted in log space, since least squares on raw singular values minimises absolute error and lets the largest value dominate a multiplicative fit.

WHAT THE RECONSTRUCTION DID NOT REPRODUCE - SINCE RESOLVED. No caption head survived, so the retrieval protocol was refitted from scratch. Measured against SigLIP 2's published figure, that first refitted head ran 7.8 points LOW. The cause turned out to be recoverable from G4's own notebook rather than an unexplained caveat, which is what made the correction below possible. The original hub was built on FOUR spaces (DINOv2 small, base and large, plus bge-m3), each scaled by its mean column standard deviation, with a RANDOM 1,000-row eval split, bge-m3 as the head's target, and ridge alpha $1e-2$. The reconstruction used all seven cached spaces, raw, with a sequential split, SBERT targets and alpha 1.0. Five differences, any of which shifts absolute retrieval.

That is the record of how the number was first obtained, and it has since been superseded. G4_cross_lineage.ipynb carries the original construction, so the protocol was rebuilt exactly rather than approximated - and it reproduces SigLIP 2's published 94.2 percent to the decimal, with DINOv2-base at 96.5 against a published 95.9 and DINOv2-large at 93.8 against 92.9. E.2's figures are therefore LEVELS, directly comparable to C.13.2, and the offset caveat is withdrawn.

Two notes survive the correction. GPT-2 is present in the seven-space reconstruction but was never in the original hub, so any statement about that concatenation's composition describes the reconstruction only. And the published singular-value table sits close to this four-space hub but not exactly on it. The best global scale is 0.9958 - essentially unity, so the construction and the scale are the same - while the shape differs by about 5 percent across widths 128 to 768, and the residual is a monotone tilt from -8.9 percent at width 64 to +3.8 percent at 768 rather than scatter. A systematic tilt of that size points to a small difference in the run that produced the table - a different row count or fit width - not a different hub. It is recorded here because it was checked, and because an earlier draft of this appendix overstated it as a different configuration on the strength of four widths rather than five. Either way the spectrum is not the gate: SigLIP's published transfer figure is, because it tests the whole pipeline including the head.

E.2 Cross-architecture transfer: a convolutional encoder

ConvNeXt-base (ImageNet-22k supervised, 1024-d) was already cached over the same 9,533 images. Row correspondence was tested rather than assumed - a ridge from DINOv2-small predicts it at R-squared 0.212 against -0.206 for a shuffled copy, a separation of 0.418 - because a misaligned encoder would yield a LOW transfer number indistinguishable from a genuine architecture effect.

Measured under the reproduced original protocol, ConvNeXt reaches 96.7 percent of a natively fitted head. That is ABOVE the 93.8 to 96.5 percent within-family band, and above SigLIP 2's 94.2 percent measured in the same run. A convolutional encoder - no attention, no patch tokenisation anywhere in it - transfers better into the hub than any transformer the band was measured on. The control sits at 0.001 against a chance rate of 0.001.

CONCLUSION. Architecture does not bound the claim, and on this evidence it does not cost anything either. Three limits belong with that. ConvNeXt is ImageNet-22k SUPERVISED where DINOv2 is self-supervised, so architecture and objective move together and neither can be isolated; since C.13.5 ranks objective above lineage above modality, objective is the more likely driver. This is one convolutional encoder on one image corpus, and a single point above a band derived from three encoders is not a trend. And the honest form of the finding remains that no encoder tested falls outside the band on the low side - not that convolutional architectures are superior.

E.3 Seven explanations for the width cliff, all falsified

C.13.11 falsified spectral amplification but bounded that falsification to one operationalisation. Seven accounts of the cliff have now been tested and all seven falsified; an eighth has survived, including a prediction made before the measurement that produced it. This section records the falsifications; E.8 records the survivor.

THE STANDARD. Log-ratio step size at the cliff relative to prior steps, on which transfer itself scores 4.03, with a pass threshold of 3.0 and a flag threshold of 2.0 calibrated against a null (0.58) and a positive control (2.33).

Re-run against the reproduced four-space hub - the object the transfer curve was actually measured on, rather than the seven-space reconstruction the first pass used - accumulated noise energy scores 0.55, the noise-to-signal direction ratio 0.39, the condition number 0.67, and the fraction of hub directions populated by all encoders 0.88. None reaches even the flag band, and none comes within a factor of four of transfer's 4.03. With local-structure loss and spectral amplification from C.13.11, that is six accounts, and every one of them failed the same way: each is SMOOTH where the outcome is discontinuous.

A note on the shared-direction figure, because it moved. The first pass computed 'populated by every encoder' over the reconstruction's seven spaces and obtained 1.00, 0.73, 0.37, 0.18, 0.12 - a steeper fall, because requiring agreement among seven encoders including three degenerate text spaces is a harder condition than the original hub ever imposed. The prediction made before re-running was that four spaces would give a higher and flatter curve, and it does: 1.00, 0.91, 0.50, 0.25, 0.17. The verdict is unchanged either way - the mechanism is real, it is smooth, and a smooth predictor cannot produce a discontinuous outcome.

THE SEVENTH ACCOUNT, falsified on different grounds. If the cliff sat at the head target's dimension rather than the source's, it would move when the target changed. G9 varied the target between bge-m3 (1024-d) and SBERT (768-d) with the source held at DINOv2-small: the cliff sat at 768 under both. The target-dimension account is the only one of the seven falsified by a wrong prediction rather than by wrong shape, which is the stronger form of refutation.

E.4 Pair ordering is not an artefact of the common-PCA reduction

C.13.7 reported Procrustes over 21 pairs after a common-PCA reduction and noted the values were not comparable to Experiment B's raw 0.038. The measurement was repeated without PCA, padding the narrower space instead.

This required a change that must be stated, because it makes the result a THIRD measure rather than a replacement for either. Orthogonal Procrustes preserves norm, and row norms across these spaces span a factor of 234 (bge 0.89, GPT-2 208.21), so rotation alone is undefined here - it produced held-out R-squared values as low as -7643, measuring scale mismatch rather than geometry. Fitting one global scale alongside the rotation restores the measure, but a similarity transform is a strictly more permissive family than rotation alone. The 0.038 and 0.194 figures are not superseded, and the C.13.7 caveat is rephrased rather than removed.

What the raw measurement establishes is the ORDERING. bge-SBERT tops the table at +0.738, image siblings follow (0.433, 0.417, 0.336), and GPT-2 pairs sit at the bottom (0.059 to 0.069) - identical to the ranking from kNN overlap, CKNN, CCA and rank correlation, now from a measure with no PCA step anywhere in it. The Table C12k ranking is not an artefact of the reduction.

With a scale fitted, the measure is also exactly symmetric: all 21 direction gaps are 0.000, so the 42-ordered-pair concern that arises under padding alone does not apply. A second quantity worth recording: the similarity transform reaches a mean of 0.281 against ridge's 0.667, so rotation and scale together recover 42 percent of what an unconstrained linear map achieves. That is a sharper statement of 'related, not rigid' than a table of failures.

E.5 The healthy-pairs null, bounded

C.13.10 reported the hub as NEUTRAL on local structure for healthy spaces, +0.010 over six pairs, and bounded that honestly as no detectable effect at this sample size rather than a demonstrated null. It can be stated more strongly without new data.

The measurement was first reproduced on the reconstructed hub. GPT-2 preservation comes out 0.262 against a published 0.257, the Spearman correlation between a pair's worst pair-cosine and its gain +0.891 against +0.820, and the degenerate-pair mean +0.122 against +0.151. The stratification holds, and SBERT again falls on the healthy side of the pair-cosine flag at 0.135 - the classification the earlier effective-rank flag got wrong.

The healthy-pair mean comes out -0.010, against the original +0.010. Two measurements under different protocols straddling zero at one hundredth is stronger evidence than either alone: the effect has no stable sign.

On independence. The six healthy pairs are every combination of four healthy encoders - DINOv2 small, base and large, and SBERT - so each encoder appears in three of them and they are not six independent observations. The interval reported here therefore resamples ENCODERS, not pairs, which widens it by a factor of 1.22 against a naive t-interval.

The cluster-bootstrap 95 percent interval is -0.038 to +0.024, against a smallest effect of interest pre-registered at 0.050 and a minimum detectable effect of 0.034. The design could have found an effect large enough to matter, and did not. This is a BOUNDED NULL rather than an absence of evidence. One caveat: with only four clusters the interval coincides with the range of the observed gains, so it is a coarse bound, not a precise estimate.

E.6 A fifth route to the local-over-global finding

One diagnostic from the reconstruction is worth recording in its own right, because it measures the project's central distinction by a route none of the earlier experiments used.

Each encoder was given one linear map into the shared hub and scored by held-out R-squared - how much of the shared coordinate system that encoder can reach from its own space. The three image encoders reach 0.46 to 0.57. All four text spaces reach between 0.02 and 0.05, and they do so uniformly, with the collapsed GPT-2 space indistinguishable from the healthy SBERT one.

The uniformity is what makes this informative rather than a restatement of the GPT-2 collapse. It is also not an alignment artefact: bge-m3's row correspondence is guaranteed by file construction rather than inferred, and bge scores 0.020, the same as the spaces whose alignment was inferred.

So the whitened hub is very largely an image-side object, and a text space sharing only a handful of directions with it can reach almost none of it. That is the same conclusion C.13.7's CCA measurement reached from cross-modal pairs sharing as few as 6 of 64 directions, and the same one C.13's composition results reached from reconstruction quality - arrived at here from the geometry of the shared coordinate system itself.

A refinement, from a later experiment. Uniform low reach does not imply the hub is insensitive to which text space it contains. G10 tests this on two axes. Holding count fixed at one text member, IDENTITY alone moves downstream transfer by 6.1 percentage points (95.3 percent bge-only to 89.2 percent bert-only) - reach cannot distinguish these four spaces, but hub composition does. Holding identity aside, PRESENCE versus absence moves transfer by 11.2 points (85.5 percent images-only to 96.7 percent seven-space). Low reach describes how much of an already-built hub a space can recover; it says nothing about how much that space changes the hub when included in its construction (E.9).

E.8 The width cliff explained: dividing by a null direction

Seven accounts of the width cliff were falsified before this one, six of them because each was smooth where the outcome is discontinuous. The eighth is confirmed, and the cliff turns out not to be a property of shared representations at all.

THE REGULARITY, PREDICTED BEFORE MEASURING. A d -dimensional source's ridge entry map is a d -by- W matrix, so its hub coordinates have algebraic rank at most d . That suggested the cliff should sit at the source encoder's own ambient dimension. The prediction was written down before running - each source cliffs at its own dimension and nowhere earlier, and the location does not move with the head target - and DINOv2-base (1536- d) and DINOv2-large (2048- d) lay outside the grid that suggested it, so both were out-of-sample tests. It held in all six curves, three sources by two targets, with the largest drop falling exactly on 768, 1536 and 2048. DINOv2-large holds above 1.03 across nine consecutive widths from 512 to 1920. Three added widths at 1024, 1280 and 1792, none of them anybody's dimension, show nothing.

THE MECHANISM, RESOLVED TO A SINGLE DIRECTION. The source's hub coordinates C have algebraic rank exactly d , so at hub width $W = d$ the LAST retained direction is the vanishing one. For DINOv2-small the singular values run 10.79 at direction 760 and 0.0054 at 768 - a ratio of two thousand. For DINOv2-base, 8.73 at 1528 and $7.6e-07$ at 1536, a ratio of twelve million. Least squares divides by the singular value, so admitting that direction gives the head enormous and meaningless weights along it - and the wider test encoders have real energy there, so it dominates their predictions.

Sweeping the retained rank in steps of eight across the boundary gives a step, not a decline. DINOv2-small holds between 0.874 and 0.894 from $k = 696$ through $k = 760$ and falls to 0.009 at $k = 768$. DINOv2-base holds between 0.959 and 0.978 from 1464 through 1528 and falls to 0.001 at 1536. The collapse coincides with one direction entering the fit, not with the accumulation of small ones - which falsifies the conditioning reading as a by-product.

Both repairs follow and both work. Truncating below d drops the direction ($k = 700$ gives 0.878 against a below-wall 0.952); ridge damps it ($\alpha 1e2$ gives 0.902, $1e3$ gives 0.917). This also explains why width 512 is safe: at $W = 512$ the coordinate matrix has full rank 512 and every retained direction is well-conditioned.

WHY EARLIER TESTS MISSED IT, RECORDED BECAUSE BOTH ERRORS WERE MINE. G13 reported the account falsified on two grounds and both were faulty. It compared an ENTROPY-BASED effective rank of 573.7 against a prediction about ALGEBRAIC rank - different quantities, and the algebraic rank is exactly 768 as predicted. And its ablation zeroed the TEST encoders' coordinates beyond direction 768 rather than removing the direction from the HEAD's fit, which is where the damage is done. G14's minimum-norm test then failed because its r_{cond} reached only $1e-4$ while the ratio that needed truncating is $0.0054/88.44$, about $6e-05$. The account was entered in the falsification ledger on the strength of those two errors and has been withdrawn from it.

WHAT THIS CHANGES IN THE BODY OF THE REPORT, and it is more than the cliff.

The width cliff is a numerical artifact of fitting a head by least squares on a rank-deficient design matrix. It is not a property of shared linear coordinates, and C.13's reading of it as the third instance of a portability-versus-losslessness tension does not survive: there is no trade-off to make, only a singular value to avoid dividing by. The meta-finding in Section 7 accordingly counts two independent instances of the isotropy trade-off, not three.

512 was never a tuned operating point. Any width strictly below the source encoder's dimension is safe, and the width-by-alpha sweep in C.13 located a rank boundary without recognising it as one. The design rule is $\text{hub_width} < \text{dim}(\text{source encoder})$, or alternatively fit the head with a truncated or heavily regularised solve and the constraint lifts entirely.

C.13's statement that alpha cannot substitute for width was measured on the ENTRY MAP alpha. The HEAD's alpha was never swept, and it is the head that breaks: at $\alpha 1e3$ transfer at width 1024 recovers to 0.917. That claim needs narrowing to what was tested.

WHAT IT DOES NOT LICENSE. Two source encoders, one hub protocol, one corpus. The mechanism is exact rather than statistical, so it should generalise wherever a head is fitted by unregularised least squares on rank-deficient coordinates - but that is an argument, not a measurement, until someone runs it elsewhere.

E.9 Sections amended by this addendum

C.13.2 - the convolutional encoder is no longer an open test; see E.2.

C.13.7 - the comparability caveat is rephrased, not removed; see E.4.

C.13 alpha claim - 'alpha cannot substitute for width' was measured on the ENTRY MAP alpha, never swept and does substitute; see E.8.

C.13.10 - the null is bounded; see E.5.

C.13.11 - the cliff is EXPLAINED. Seven accounts falsified, the eighth confirmed to single-direction resolution on two encoders; see E.3 and E.8.

Section 7.4 falsification ledger - eleven entries, two of them this project's own. The eleventh is a later addition, qualifying E.6; see E.6 and Results_Summary. The rank account was entered prematurely on a faulty comparison and has been withdrawn; its place is taken by the conditioning reading, which the single-direction step falsifies.

Section 7 meta-finding - the cliff is a numerical artifact and is NOT a third instance of the isotropy trade-off; two instances, not three; see E.8.

C.13 operating point - 512 is safe because it is below the source encoder's rank, not because it is optimal. The constraint lifts entirely under a truncated or heavily regularised head; see E.8.

Nothing in the body is retracted. Every change is an addition or a narrowing of a stated boundary.

One item remains genuinely open and is recorded here rather than left silent: benchmark B1, comparing an ensemble of specialists against one large model, was pre-registered in Sources_and_Benchmarks and has not been run. It is deferred for time rather than abandoned, and nothing elsewhere in this report depends on its outcome.

Six sections are ADDED rather than amended. E.10 records each encoder's training objective and supervision regime. E.11 positions Experiments B and G against the model-stitching literature. E.12 extends the shape-agreement matrix to ConvNeXt. E.13 records the rows-per-dimension of each hub entry map and recovers the native R@1 that Table C12 leaves implicit. E.14 corrects the drafting error of applying the regression floor to the hub's PCA, and separates C.11's full-rank whitening from the hub's truncate-then-whiten. E.15 records that row correspondence across caches is positional - two `keep` conventions coexist and cannot be joined by id - and that it is verified by shuffle test.

E.10 Encoder provenance: objective, supervision regime, and role

The body of this report names every encoder by architecture and never records how it was TRAINED. That omission matters for reading the convergence results, because the roster is not one supervision regime but three. This section states, for each space, its training objective, what supervises it, whether it ever saw text, and the measurement it contributes.

encoder / dim	objective and what supervises it	role and measured result
DINOv2 small / base / large 768 / 1,536 / 2,048	Self-distillation - DINO at image level, iBOT at patch level. The targets are an EMA teacher's outputs on OTHER crops of the SAME image, so the signal is manufactured by training rather than annotated, and no text enters the loss. LVD-142M was retrieval-curated against curated seed collections, so human judgment enters at the CORPUS level, not per example - self-supervised on automatically curated data, not unsupervised.	Hub members and the capacity ladder: 40.9 / 53.5 / 58.3 percent of ceiling into bge, +15.3 points per decade of parameters, fit R2 0.949. img_small is the narrowest source in the set and therefore sets the hub's usable width, hub_width < 768 (E.8).
bge-m3 1,024	Contrastive on mined and curated query-passage pairs. PAIR-SUPERVISED - the supervision is the pairing, not a class label. Text only.	Hub member and the head's target in the published protocol. Half of the top pair with SBERT at +0.738 raw Procrustes: shared objective, no shared lineage (E.4).
GPT-2 768	Next-token prediction. SELF-SUPERVISED, no annotation - the same hold-out-and-predict construction as DINOv2, on tokens rather than pixels.	The collapse case: effective rank 6.6, read out of at 95.1 percent but written into at only 24.8. Whitening lifts R@1 from 0.130 to 0.478 = 102.6 percent of an encoder trained contrastively on 1.2 billion pairs (C.11).
BERT 768	Masked language modelling. SELF-SUPERVISED, no annotation.	Hub member and a rung of the encoder ladder; effective rank 65.0, between GPT-2 and the contrastively trained text spaces.
SBERT 768	BERT fine-tuned on NLI - human-labelled entailment pairs. SUPERVISED in the strict sense, and the only text space here that is.	Best head target at 101.0 percent of native; healthy on the pair-cosine flag at 0.135 (K2); the other half of the top pair with bge-m3.
SigLIP 2 768	Image-text contrastive on web alt-text. PAIR-SUPERVISED, and one of only two image encoders in the project that see text at all.	HELD OUT. Cross-lineage transfer 94.2 percent, inside the 93.8-96.5 within-family band and reproduced to the decimal by G0-exact.
ConvNeXt-base 1,024	ImageNet-22k classification. FULLY SUPERVISED on 22,000 human categories, and a convnet rather than a transformer - two axes of difference at once.	HELD OUT. 96.7 percent of native, ABOVE the within-family band and above SigLIP 2 in the same run (E.2). Neither architecture nor supervision regime bounds the claim.
MobileCLIP-S1 512	Image-text contrastive, distilled. PAIR-SUPERVISED.	Experiment B source: ridge adapter R2 0.592, mean cosine 0.900, retrieval 88.4 percent of ceiling; shipped at 0.79 MB as Core ML fp16.

What the mix buys, and two boundaries it does not lift. Encoders that never saw a label, encoders trained on web pairings, and a classifier trained on 22,000 human categories all join one hub and carry a head at 93 to 97 percent of native, so convergence does not require a shared supervision signal - which is what keeps the thesis from being circular. G5 measured the regime's role directly: objective explains pairwise agreement more than lineage or modality, +53x against +44x and +29x, making supervision a modulator of HOW MUCH encoders agree rather than a precondition for WHETHER they do. Two boundaries follow and are not lifted by any of it. C.12's pair-free recovery ran on two contrastive image-text encoders - the weakest-independence pair in the project - and is bounded accordingly. And the label-free claim is about the OBJECTIVE: DINOv2's corpus was shaped by retrieval against curated collections, so no training example carries an annotation while the distribution itself was not judgment-free.

E.11 Stitching: what B and G share with it, and where the analogy stops

Model stitching is the label a reader is most likely to reach for here, and for Experiment B it is the right one. Setting out what the two experiments share with it, and where the analogy stops, is more useful than resisting the comparison.

	classic stitching	Experiment B	Experiment G (the hub)
what is frozen	both networks; only the connector is free	MobileCLIP-S1 and SigLIP 2, both frozen	all seven encoders; the entry maps and one head are the only fitted objects
what the connector is fitted ON	the downstream LOSS, by SGD - so it can learn to compensate for a mismatch	representation matching, closed-form ridge. Never sees the retrieval objective	the same: closed-form ridge into a whitened PCA basis. No task signal reaches it
where the join sits	an intermediate layer; results are strongly depth-dependent	the pooled output embedding only	the pooled output embedding only
maps needed for N encoders	one per ordered pair, so $N(N-1)$	one, for one ordered pair	N entry maps into one pivot - seven rather than forty-two, and a new encoder leaves every existing map untouched
an encoder unseen at fitting time	impossible: the connector is defined per pair	impossible, for the same reason	supported. SigLIP 2 at 94.2 percent of native, ConvNeXt-base at 96.7, neither present when hub or head was fitted
measured cost	small, in the original papers	R2 0.592, mean cosine 0.900, 88.4 percent of the retrieval ceiling; ships at 0.79 MB	a head transfers at 92.9-96.5 percent of native; composition retains 95.6 percent of directly fitted maps over 36 of 41 ordered pairs
direction	one way, A into B	one way	read-out and write-in differ: GPT-2 reads out at 95.1 percent and is written into at 24.8 (C.13)

Two baselines, answering different questions. G1 compares composition through the hub against a ridge map fitted DIRECTLY for that pair: 87.0 percent retention over all 41 ordered pairs, 95.6 over the 36 that do not write into GPT-2 (Table C12c). G3 compares the shared hub against BESPOKE PER-PAIR HUBS, which is the fair baseline for the pivot claim rather than for the connector claim, and finds 42 of 42 ordered pairs shared at a mean retention of 103.6 percent, control at chance, 0.004 against 0.001. Sharing costs nothing and marginally gains: the pooled basis is estimated from seven spaces rather than two. The two pair counts, 41 and 42, come from different protocols and are quoted as each was run. **What this positions the project as.** B is stitching - at the output layer, with a connector that never sees the task, which makes its 88.4 percent a stronger statement than a connector trained against the objective it is scored on. G is a pivoted generalisation with no pairwise analogue, because a stitching connector to an encoder that was absent when the connector was fitted is not a thing that can exist. The stitching literature asks whether a connector CAN be found; this project asks what the connector runs on, and answers with local structure over global (four metrics, C.13.10), objective over lineage over modality (G5), and a bounded null on the hub's cost (E.5). One boundary in the other direction: stitching results are depth-dependent and these are not, since every join here is at the pooled output. And the width cliff of E.8 belongs to this literature rather than to the representations - it is a property of FITTING the connector by least squares on rank-deficient coordinates, not of the spaces being connected.

E.12 ConvNeXt's raw geometry: what the hub recovers that shape agreement does not show

E.2 measured ConvNeXt-base at 96.7 percent of native as a zero-shot source, above the within-family band. What was never measured is its RAW shape agreement - the rank correlation between its pairwise-distance structure and every other encoder's, before any hub exists and with no map fitted. Section C.13.9's matrix covered the seven hub members only, because ConvNeXt was deliberately held out of the hub. Extending that matrix to eight encoders closes the gap, and the answer is not the expected one.

ConvNeXt-base against	rank correlation (rho)	for comparison
DINOv2-small	0.397	DINOv2 variants agree with EACH OTHER at 0.56 to 0.60
DINOv2-base	0.392	
DINOv2-large	0.391	
SBERT	0.330	ConvNeXt's best text pair
BERT	0.305	DINOv2-large's best text pair is BERT at 0.418
bge-m3	0.302	
GPT-2	0.166	the collapsed space; lowest for every encoder

Spearman of pairwise cosine distances, all 9,533 items, no map fitted. Table C12m.

The finding, and it inverts the expected direction. ConvNeXt has the LOWEST raw shape agreement of any image encoder in the roster. Its best cross-modal pair reaches 0.330, below DINOv2-large's 0.418; and even its agreement with the other IMAGE encoders sits at 0.39 to 0.40, well below the 0.56 to 0.60 the three DINOv2 variants reach with one another. By raw geometry ConvNeXt is the outsider of the set, which is what its provenance predicts: a convnet trained on 22,000 human labels, differing from every other encoder on architecture AND supervision at once. And yet it is the BEST-transferring encoder measured - 96.7 percent of native in the four-space hub, 99.3 percent in the seven-space one (G10), above the entire DINOv2 family whose raw geometry it agrees with least.

What this establishes. The 96.7 percent is not underwritten by superior raw geometric agreement; it is achieved DESPITE the weakest raw agreement in the set. The hub and its entry map are therefore recovering a linear correspondence that the shape-agreement metric does not register - the two quantities are measuring different things, and one does not predict the other. This is the second independent instance of the lesson E.6 records for text spaces, where uniformly low reach (R-squared 0.02 to 0.05) failed to predict an 11.2-point spread in downstream transfer. That instance varied the text side of a fixed hub; this one varies the image side, with an encoder held out of the hub entirely. The principle generalises: a raw-geometry diagnostic bounds what is VISIBLE without fitting, not what is RECOVERABLE with one closed-form solve.

What it does not establish. One encoder is a demonstration, not a law - the inversion is shown, not shown to be general. The two measurements are also not commensurable in the strict sense: shape agreement fits nothing, while transfer fits an entry map on training rows, so the comparison is between what raw geometry reveals and what a fitted map achieves, not between two estimates of one quantity. Nothing here revises E.2's figure or the hub protocol; it removes an assumption that was never tested - that hub transfer tracks raw agreement - and records that it does not.

E.13 Statistical power of the hub entry maps, and the native reference cross-check

Two bookkeeping facts that Table C12 does not state, and one an examiner can derive. Table C12 reports R@1 and percent of native for the three DINOv2 rungs. Neither the R@1 of the native head itself nor the rows-per-dimension of each entry map appears there, and both bear on how the percentages should be read.

How the corpus size was set, and why it does not carry into the hub. The E-series sizes each experiment from the encoder's own width: $N = 1,000 + 10 \times \text{width} / 0.9$, so the training rows are $10 \times \text{width} / 0.9$ and the width cancels, giving 11.1 rows per input dimension for every rung. The 10 is the target, twice the project's floor of 5; the 0.9 over-requests by about eleven percent so that expected fetch losses still leave ten. In the event no images were lost in any of the three runs - 9,533, 18,066 and 23,755 requested, the same number cached - so the realised rate equalled the planned 11.1 exactly. The HUB, however, is fixed at 9,533 rows, which is DINOv2-small's corpus, because the concatenation requires the same item in every space and the intersection is bounded by the narrowest cache. Wider encoders therefore enter the hub with fewer rows per dimension than their own experiment gave them.

encoder / width	own experiment	as a hub entry map	consequence
DINOv2-small 768	11.1 rows/dim	11.1 rows/dim	the hub corpus IS its corpus; no loss
DINOv2-base 1536	11.1 rows/dim	5.6 rows/dim	above the floor of 5
DINOv2-large 2048	11.1 rows/dim	4.2 rows/dim	BELOW the floor; ridge biases pessimistic, so 92.9 percent is conservative
ConvNeXt-base 1024	held out	8.3 rows/dim	above the floor

Entry maps are fitted on the 8,533 training rows; rows/dim is 8,533 divided by the encoder's width. Table C12n.

The native reference, recovered, and an unplanned consistency check. Percent of native is R@1 transferred divided by R@1 of a head fitted on that encoder's own hub coordinates. Inverting it recovers the native figure Table C12 leaves implicit. The recovered values then agree with E1's DIRECT image-to-bge ridge map, a separate experiment on a separate protocol, to three decimals on two of three rungs.

encoder	R@1 transferred	percent of native	R@1 native, recovered - and E1's direct map
DINOv2-small	0.398	111.5	$0.398 / 1.115 = 0.357$ E1 direct: 0.357
DINOv2-base	0.442	95.9	$0.442 / 0.959 = 0.461$ E1 direct: 0.466
DINOv2-large	0.468	92.9	$0.468 / 0.929 = 0.504$ E1 direct: 0.504

E1 direct-map figures are E1_REF as recorded in the E1.1 notebook. Table C12o.

What both tables change in how C12 should be read. First, absolute retrieval RISES monotonically with encoder capacity - 0.398, 0.442, 0.468 - while the percentage FALLS. The percentage falls because its denominator rises faster: a larger encoder supports a better native head, so the same transferred head recovers a smaller share of a higher ceiling. The falling percentage is not falling performance, and should not be read as one. Second, the hub costs nothing at the reference point: a head fitted natively on hub coordinates matches a direct ridge into bge-m3 to three decimals on two rungs and within 0.005 on the third, from two experiments that share no fitting code. Third, DINOv2-large's entry map is the one fit in the published protocol that sits below the project's own power floor, at 4.2 rows per dimension; the direction of that bias is downward, so 92.9 percent is if anything an underestimate. It is recorded here rather than left for a reader to derive.

E.14 The hub basis: aspect ratio, and a floor that does not apply to it

Why the project's power floor is the wrong instrument here. E.13's entry maps are regression fits, and this project's floor of five rows per input dimension is a regression rule: it counts observations per estimated coefficient. The hub BASIS is not a regression. It is a principal-component decomposition of the concatenated spaces, and the governing quantity is the aspect ratio $\gamma = d/n$. Applying the regression floor to a PCA overstates the problem, and it is recorded here as a correction rather than left for a reader to make.

hub configuration	concatenated d	$\gamma = d / 8,533$	rank
4-space, the published protocol	5,376	0.63	full ($n > d$)
5-space, Section C.13 body	6,144	0.72	full ($n > d$)
7-space, Section C.13.3 / G1	7,680	0.90	full, but only just

Concatenation width is the sum of member widths: 768 + 1536 + 2048 + 1024 (+ 768 GPT-2) (+ 768 BERT + 768 SBERT). Table C12p.

What gamma at 0.90 does and does not imply. The covariance stays full rank in every configuration, since n exceeds d throughout. But as γ approaches one the sample eigenvalue spectrum spreads away from the population spectrum, so the basis is estimated less precisely than a low- γ fit would give. Two considerations bound the concern. Only the leading 512 directions are retained, about seven percent of the 7,680-dimensional spectrum, and leading eigenvectors are recovered far more reliably than the bulk - so the retained subspace is the best-estimated part of what was computed. And the residual distortion degrades transfer rather than inflating it, placing every figure in this report on the conservative side. A 35,133-item corpus would bring γ to 0.22; that is recorded as future work, not claimed as done.

One encoder that does not affect it. ConvNeXt-base is not a hub member. Its 1,024 dimensions never enter the concatenation, so it does not move γ ; had it been included, the seven-space figure would have risen from 0.90 to 1.02 and the covariance would have become rank-deficient. ConvNeXt instead receives only an entry map, fitted at 8.3 rows per dimension. This is the property that makes the hub extensible: encoders added after construction cost one closed-form solve and leave the basis untouched.

A second question the aspect ratio raises: where the whitening is applied. The hub whitens, and Section C.11 whitens, but they are not the same operation and the second does not reach the first. C.11 takes GPT-2's own 768-dimensional space and whitens it IN FULL - every direction rescaled, nothing discarded - lifting its effective rank from 6.5 to 490.8 and its $R@1$ from 0.130 to 0.478. The hub does something different in a different order: each space is first rescaled by a single scalar, the mean of its per-column standard deviations, which equalises scale ACROSS spaces while leaving each one's internal geometry untouched; the spaces are concatenated; the top 512 directions of the joint spectrum are retained; and only those survivors are divided by their singular values.

operation	applied to	truncation	effect on GPT-2
C.11 / E1.2 rescue	GPT-2's own 768-d space	none	rank 6.5 -> 490.8; $R@1$ 0.130 -> 0.478
hub standardisation	each space separately	none	one scalar; collapse preserved intact
hub whitened PCA	the joint 5,376-d concat	top 512 kept	equalises the survivors only

Source: G0_hub_exact and G1_shared_hub for the hub; E1_2_isotropy_rescue for the C.11 transform. Table C12q.

What follows, and what does not. Because truncation precedes whitening, the directions the hub equalises are the ones that already carried variance. GPT-2's effective rank is 6.5, so it contributes roughly six high-variance directions to the joint spectrum and those survive the cut; the remaining seven hundred and sixty - precisely the directions C.11's full-rank whitening lifted - sit in the tail and are discarded before any whitening reaches them. G3's note that the hub's whitening rescues a collapsed space is therefore correct about what the hub RETAINS, and not in tension with this: the two transforms simply act on different sets of directions. The scalar rescale is also the right choice for its own purpose, and deliberately so - per-column standardisation would inflate GPT-2's near-dead columns to unit variance before the PCA ever saw them, which is the failure mode the width cliff of E.8 describes. What remains untested is the third option: apply C.11's full-rank whitening to GPT-2 BEFORE it enters the concatenation, and rebuild. G10 already shows GPT-2's presence in the hub is worth several points; whether a rescued GPT-2 is worth more is a one-line change and has not been run.

E.15 Row correspondence across caches: positional, verified by shuffle, never by id

How rows are known to correspond, and how they are not. Every number above assumes row i of one cache is the same COCO item as row i of every other. That holds, but not for the reason a reader might expect. Two conventions for the per-row `keep` field coexist in the project: the E1.1-era caches (DINOv2 small, base, large - and `hub_ids` in `hub_rebuilt.npz`, which G0 derives from them) record the request POSITION, 0 to N-1, carrying no image identity; the G4-era caches (SigLIP 2, ConvNeXt) record the real COCO image id. An id join across the two groups is therefore meaningless - the 1,879 ids they appear to share is simply the number of COCO ids that fall below 9,533, a coincidence G4's ConvNeXt notebook already records. Correspondence holds because every cache was written by the same deterministic loop over the same sorted id list, and it is verified the only way that applies across namespaces: a ridge or rank correlation between two caches must exceed the same statistic under a row shuffle by a wide margin. The scale extension reproduces that check for each new encoder (gaps of 0.26 to 0.85 against a shuffled 0.00).

cache generation	what <code>keep</code> stores	joinable by id?
E1.1: DINOv2 s/b/l; <code>hub_ids</code>	request position 0..N-1 (arange)	no - carries no identity
G4: SigLIP 2, ConvNeXt	real COCO image id (9..46492)	only with each other
scale extension: giant, Qwen	request position, as E1.1	no - same as E1.1

Alignment is by shared deterministic ordering, verified by shuffle test - never by id. Table C12r.

How the corpus size propagates, and why every later encoder is also 9,533. The two loops select their items with the identical expression, which is what makes the orderings agree in the first place:

```
E1.1 and the scale extension
ids = sorted(set(caps) & set(url))[:N_PAIRS]
G4_cross_lineage (SigLIP 2, ConvNeXt)
N_PAIRS = min(len(np.load(f)["img"]) for f in dinov2_caches)
ids      = sorted(set(caps) & set(url))[:N_PAIRS]
```

The consequence. G4 does not choose a corpus size at all: it takes the row count of the NARROWEST existing DINOv2 cache and truncates to that. Since DINOv2-small's own experiment is 9,533 rows, every encoder added afterwards inherits 9,533 - SigLIP 2, ConvNeXt, and the scale extension's DINOv2-giant and Qwen models alike. This is the mechanism behind the statement in E.13 that the hub corpus is bounded by its smallest member: the bound is not merely a property of the intersection, it is written into the loader, so it propagates automatically to encoders added years after the hub was built. An encoder whose own E-series rule would call for 35,133 rows, as DINOv2-giant at `cls+patch` does, is still cached at 9,533 - which is correct for hub work, where every space must describe the same items, and is why its entry map sits at 2.8 rows per dimension and its transfer figure is reported as inconclusive rather than as a capacity result.